

Three-dimensional Geometry

Version 1.0

20-12-2020

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Introduction

Three-dimensional geometry used to be a topic in the IMO, but has disappeared for many years. Therefore, this note is mainly for fun.

It is not easy to visualize a three-dimensional figure on a two-dimensional paper. So three-dimensional geometry problems are normally much harder. It is necessary to go through some standard proofs of common results to see what are the usual techniques involved in solving three-dimensional geometry problems. This note will cover some of these important tools. It is also interesting to take a look at those well-known results, and compare with those results in two-dimensional geometry.

In section 1, we introduce the three-dimensional space using **vectors**. The approach involves a bit of algebra, but this is unavoidable if one wants to explore higher dimensional spaces. In fact, to study general n -dimensional spaces, we can only rely on algebra because those spaces cannot be visualized. In the second part, we give a brief introduction to the common geometric objects in the three-dimensional space. There are more detailed discussions on planes and **spheres** because they are some important objects.

In section 2, we go through different centres of **tetrahedrons** analogous to those centres of triangles. These include the **centroid**, the **circumcentre**, the **incentre** and the **excentres**. We will also introduce the **Monge point** and the **twelve-point centre**.

In section 3, we talk about a special kind of **tetrahedron** called **orthocentric tetrahedrons**. It is the class of **tetrahedron** which has an **orthocentre**. For general **tetrahedrons**, the **orthocentre** needs not exist. We will see what are the conditions for a **tetrahedron** to have an **orthocentre**.

1 Vectors and the Three-dimensional Space

In this section, we use **vectors** to introduce the three-dimensional space, since this is a more rigorous way to define the **Euclidean space** and can be easily generalized to higher dimensional cases. However, in most parts of the remaining sections, we will not use **vectors**. Therefore, it is fine to skip the first subsection. Subsection 1.2 is to introduce various geometric objects in the three-dimensional space, some of which will be used later.

1.1 Vectors

Definition 1.1. A *vector*(向量) is an n -tuple $(v_1, v_2, \dots, v_n)$ of real numbers. The set $\mathbb{R}^n$ of all such **vectors** is called the *Euclidean space*(歐幾里得空間) of dimension n .

In particular, the usual plane geometry refers to geometry in $\mathbb{R}^2$. In this note, our main theme is the **Euclidean space** $\mathbb{R}^3 = \{(x, y, z) : x, y, z \in \mathbb{R}\}$. From now on, we restrict our attention to $\mathbb{R}^3$, but many definitions in this subsection can be generalized to arbitrary $\mathbb{R}^n$.

Two **vectors** $\mathbf{u} = (u_1, u_2, u_3)$ and $\mathbf{v} = (v_1, v_2, v_3)$ are said to be equal if $u_j = v_j$ for each $j = 1, 2, 3$. We can define

$$\mathbf{u} \pm \mathbf{v} = (u_1 \pm v_1, u_2 \pm v_2, u_3 \pm v_3) \quad \text{and} \quad c\mathbf{v} = (cv_1, cv_2, cv_3)$$

where $c \in \mathbb{R}$. The product $c\mathbf{v}$ is called the *scalar multiplication*(純量乘法), while the real number c is called a *scalar*(純量) so as to distinguish it from a **vector**. These basic operations come from the usual binary operations on real numbers, so they have all the nice properties of addition, subtraction and multiplication. The **vector** $\mathbf{0} = (0, 0, 0)$ is called the *zero vector*(零向量).

Apart from understanding **vectors** algebraically, we can also visualize **vectors** in a geometric way. Note that we can easily generalize the Cartesian coordinate system in $\mathbb{R}^2$ to higher dimensional cases, so that each point in $\mathbb{R}^n$ is represented by the coordinates $(x_1, x_2, \dots, x_n)$.

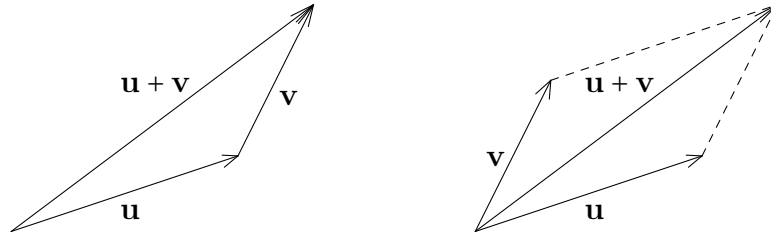
Definition 1.2. Let P be a point in $\mathbb{R}^3$ with coordinates (x, y, z) . The *position vector*(位置向量) of P is an arrow drawing from the origin $O(0, 0, 0)$ to the point P , and is denoted by $\overrightarrow{OP}$.

This gives us the connection between the algebra and geometry of **vectors**. The **vector** $\mathbf{v} = (x, y, z)$ can be visualized as the **position vector** of (x, y, z) . In general, we

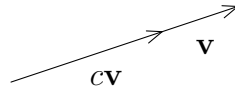
can use an arrow of the same length and direction to represent the same **vector**, not necessarily having O as the initial point. For example, we can draw an arrow from the point $(1, 2, 3)$ to the point $(4, 6, 8)$ to represent the **vector** $(4, 6, 8) - (1, 2, 3) = (3, 4, 5)$.

The *standard vectors*(標準向量) are the three **vectors** $\mathbf{i} = (1, 0, 0)$, $\mathbf{j} = (0, 1, 0)$ and $\mathbf{k} = (0, 0, 1)$ pointing in the positive x , y and z -directions respectively. Using these notations, we can also write the **vector** $\mathbf{v} = (x, y, z)$ as $\mathbf{v} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$.

The arithmetic operations on **vectors** can be interpreted in a geometric way. If we put the tail of $\mathbf{v}$ at the tip of $\mathbf{u}$, then the **vector** $\mathbf{u} + \mathbf{v}$ is represented by the arrow drawing from the tail of $\mathbf{u}$ to the tip of $\mathbf{v}$. Alternatively, if we put the tails of $\mathbf{u}$ and $\mathbf{v}$ at the same point and draw a parallelogram using $\mathbf{u}$ and $\mathbf{v}$ as two of the sides, then the **vector** $\mathbf{u} + \mathbf{v}$ is represented by the arrow drawing from the tail of $\mathbf{u}$ to the opposite vertex of the parallelogram.



For **scalar multiplication**, $\mathbf{v}$ and $c\mathbf{v}$ are parallel **vectors**. When $c > 0$, the **vector** $c\mathbf{v}$ has the same direction as $\mathbf{v}$ but its length is c times that of $\mathbf{v}$. When $c = 0$, the **vector** $c\mathbf{v}$ is the **zero vector**. When $c < 0$, the **vector** $c\mathbf{v}$ has opposite direction as $\mathbf{v}$ and its length is $|c|$ times that of $\mathbf{v}$. In particular, $-\mathbf{v}$ has the same length as $\mathbf{v}$, but points in the opposite direction as $\mathbf{v}$. Using this, it is not hard to find some geometric meaning of $\mathbf{u} - \mathbf{v} = \mathbf{u} + (-\mathbf{v})$.



Using Pythagoras' theorem, it is natural to define the length of a **vector** $\mathbf{v} = (x, y, z)$ as $\sqrt{x^2 + y^2 + z^2}$. We also call this the *norm*(範數) of $\mathbf{v}$ and use the notation $\|\mathbf{v}\|$. A *unit vector*(單位向量) is a **vector** with **norm** 1.

Next, we introduce some useful products between **vectors**.

Definition 1.3. The *dot product*(點積) of $\mathbf{u} = (u_1, u_2, u_3)$ and $\mathbf{v} = (v_1, v_2, v_3)$ is defined by

$$\mathbf{u} \cdot \mathbf{v} = u_1v_1 + u_2v_2 + u_3v_3.$$

Like the usual product, the **dot product** satisfies the following nice properties.

Proposition 1.1. Let $\mathbf{u}, \mathbf{v}$ and $\mathbf{w}$ be any **vectors** in $\mathbb{R}^3$, and let a be any real number. Then the following hold.

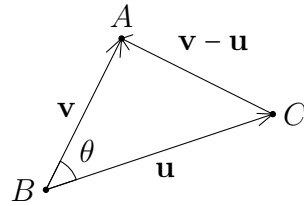
- (a) $(\mathbf{u} + \mathbf{v}) \cdot \mathbf{w} = \mathbf{u} \cdot \mathbf{w} + \mathbf{v} \cdot \mathbf{w}$
- (b) $(a\mathbf{u}) \cdot \mathbf{v} = a(\mathbf{u} \cdot \mathbf{v})$
- (c) $\mathbf{u} \cdot \mathbf{v} = \mathbf{v} \cdot \mathbf{u}$
- (d) $\mathbf{v} \cdot \mathbf{v} = \|\mathbf{v}\|^2 \geq 0$, and equality holds if and only if (iff) $\mathbf{v} = \mathbf{0}$

The most important property of the **dot product** is its relation with angles.

Proposition 1.2. Let $\mathbf{u}$ and $\mathbf{v}$ be any nonzero **vectors** in $\mathbb{R}^3$. Let θ be the angle between $\mathbf{u}$ and $\mathbf{v}$. Then

$$\mathbf{u} \cdot \mathbf{v} = \|\mathbf{u}\| \|\mathbf{v}\| \cos \theta.$$

In particular, $\mathbf{u}$ and $\mathbf{v}$ are perpendicular iff $\mathbf{u} \cdot \mathbf{v} = 0$.



Proof. We put the **vectors** on a plane in the way such that they have the same tail B . Suppose $\mathbf{u} = \overrightarrow{BC}$ and $\mathbf{v} = \overrightarrow{BA}$. The angle θ between $\mathbf{u}$ and $\mathbf{v}$ is defined as $\angle CBA$. Note that $\overrightarrow{CA} = \overrightarrow{CB} + \overrightarrow{BA} = -\mathbf{u} + \mathbf{v}$. We apply the cosine formula in $\triangle ABC$. Now,

$$\begin{aligned} AC^2 &= AB^2 + BC^2 - 2AB \cdot BC \cos \theta \\ \Rightarrow \quad \|\mathbf{v} - \mathbf{u}\|^2 &= \|\mathbf{v}\|^2 + \|\mathbf{u}\|^2 - 2\|\mathbf{v}\|\|\mathbf{u}\| \cos \theta \\ \Rightarrow \quad (\mathbf{v} - \mathbf{u}) \cdot (\mathbf{v} - \mathbf{u}) &= \mathbf{v} \cdot \mathbf{v} + \mathbf{u} \cdot \mathbf{u} - 2\|\mathbf{v}\|\|\mathbf{u}\| \cos \theta \\ \Rightarrow \quad \mathbf{u} \cdot \mathbf{u} - 2\mathbf{u} \cdot \mathbf{v} + \mathbf{v} \cdot \mathbf{v} &= \mathbf{v} \cdot \mathbf{v} + \mathbf{u} \cdot \mathbf{u} - 2\|\mathbf{v}\|\|\mathbf{u}\| \cos \theta \\ \Rightarrow \quad \mathbf{u} \cdot \mathbf{v} &= \|\mathbf{u}\| \|\mathbf{v}\| \cos \theta. \end{aligned}$$

This proves the proposition. □

Indeed, rather than proving this result using the cosine formula (which is a well-known fact to us in $\mathbb{R}^2$), it is more common in mathematics to use this as the definition of the angle between two **vectors**.

Theorem 1.1. (Cauchy-Schwarz inequality(柯西不等式)) Let $\mathbf{u}$ and $\mathbf{v}$ be any **vectors** in $\mathbb{R}^3$. Then

$$|\mathbf{u} \cdot \mathbf{v}| \leq \|\mathbf{u}\| \|\mathbf{v}\|.$$

Proof. This follows from proposition 1.2 since

$$|\mathbf{u} \cdot \mathbf{v}| = \|\mathbf{u}\| \|\mathbf{v}\| |\cos \theta| \leq \|\mathbf{u}\| \|\mathbf{v}\|$$

where θ is the angle between $\mathbf{u}$ and $\mathbf{v}$. □

We will not use this inequality in this note. This is just to mention this **vector** form of the **Cauchy-Schwarz inequality**. Indeed, the same result holds in $\mathbb{R}^n$. If we write $\mathbf{u} = (u_1, u_2, \dots, u_n)$ and $\mathbf{v} = (v_1, v_2, \dots, v_n)$, then the above inequality becomes

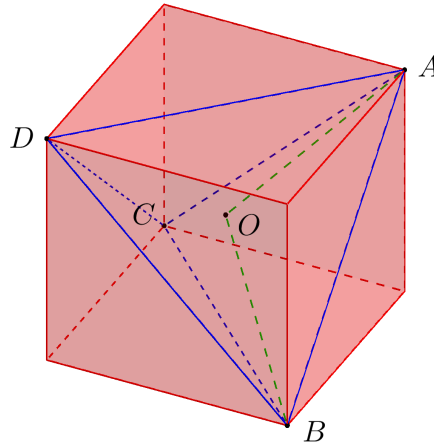
$$|u_1 v_1 + u_2 v_2 + \dots + u_n v_n| \leq \sqrt{u_1^2 + u_2^2 + \dots + u_n^2} \sqrt{v_1^2 + v_2^2 + \dots + v_n^2},$$

which is the usual form of the **Cauchy-Schwarz inequality**.

Problem 1.1. Let O be the centre of a **regular tetrahedron** $ABCD$. Find $\angle AOB$.

Analysis. This example shows how we can compute some angles in space using **dot product**. We first need to put the **tetrahedron** in a fixed position, and then find $\vec{OA}$ and $\vec{OB}$. To do so, we find that the best way is to pick out the **tetrahedron** from four non-adjacent vertices of a **cube**. The definitions of **tetrahedrons** and **cubes** will be given in subsection 1.2.

Solution.



Consider a **cube** of length 2 with centre O . Let $A(1, 1, 1)$, $B(1, -1, -1)$, $C(-1, 1, -1)$ and $D(-1, -1, 1)$ be four of the vertices which are pairwise non-adjacent. By symmetry, $ABCD$ is a **regular tetrahedron** with centre O . We now apply proposition 1.2.

Note that

$$\begin{aligned}
& \vec{OA} \cdot \vec{OB} = \|\vec{OA}\| \|\vec{OB}\| \cos \angle AOB \\
\Rightarrow & (1, 1, 1) \cdot (1, -1, -1) = \|(1, 1, 1)\| \|(1, -1, -1)\| \cos \angle AOB \\
\Rightarrow & -1 = (\sqrt{3})(\sqrt{3}) \cos \angle AOB \\
\Rightarrow & \cos \angle AOB = -\frac{1}{3} \\
\Rightarrow & \angle AOB = \cos^{-1}\left(-\frac{1}{3}\right) \approx 109.47^\circ.
\end{aligned}$$

Problem 1.2. (Sharygin Geometry Olympiad 2011 Final Grade 10 Problem 3) Given two **tetrahedrons** $A_1A_2A_3A_4$ and $B_1B_2B_3B_4$, consider six pairs of edges A_iA_j and B_kB_ℓ , where (i, j, k, ℓ) is a transposition of numbers $(1, 2, 3, 4)$ (for example A_1A_2 and B_3B_4). It is known that for all but one such pairs the edges are perpendicular. Prove that the edges in the remaining pair are also perpendicular.

Analysis. We basically do not need any facts about **tetrahedrons**. The only trick is to rewrite the perpendicular conditions as some **dot products**, and work on the algebra.

Solution. Let $\mathbf{a}_i$ and $\mathbf{b}_j$ be the **position vectors** of A_i and B_j respectively. Without loss of generality (WLOG), assume all pairs of edges except A_3A_4 and B_1B_2 are perpendicular. Then the perpendicular conditions imply

$$(\mathbf{a}_2 - \mathbf{a}_1) \cdot (\mathbf{b}_4 - \mathbf{b}_3) = 0, \quad (1)$$

$$(\mathbf{a}_3 - \mathbf{a}_1) \cdot (\mathbf{b}_4 - \mathbf{b}_2) = 0, \quad (2)$$

$$(\mathbf{a}_4 - \mathbf{a}_1) \cdot (\mathbf{b}_3 - \mathbf{b}_2) = 0, \quad (3)$$

$$(\mathbf{a}_3 - \mathbf{a}_2) \cdot (\mathbf{b}_4 - \mathbf{b}_1) = 0, \quad (4)$$

$$(\mathbf{a}_4 - \mathbf{a}_2) \cdot (\mathbf{b}_3 - \mathbf{b}_1) = 0. \quad (5)$$

Now, by simplifying $-(1)+(2)-(3)-(4)+(5)$, we obtain

$$(\mathbf{a}_4 - \mathbf{a}_3) \cdot (\mathbf{b}_2 - \mathbf{b}_1) = 0.$$

This implies $A_3A_4 \perp B_1B_2$ as desired.

Remark. It is not hard to see how the equations can be combined. For example, to obtain the term $\mathbf{a}_4 \cdot \mathbf{b}_2$ at the end, it must come from (3) with a negative sign, etc.

Definition 1.4. The *cross product* (叉積) of $\mathbf{u} = (u_1, u_2, u_3)$ and $\mathbf{v} = (v_1, v_2, v_3)$ is defined by

$$\mathbf{u} \times \mathbf{v} = (u_2v_3 - u_3v_2, u_3v_1 - u_1v_3, u_1v_2 - u_2v_1) = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{vmatrix}.$$

The matrix determinant notation is just a convenient way to express the formula. But it is not well-defined since we normally do not put different things (**vectors** and real numbers) as the entries of a matrix. Note that the **cross product** can only be defined in $\mathbb{R}^3$.

Proposition 1.3. Let $\mathbf{u}, \mathbf{v}$ and $\mathbf{w}$ be any **vectors** in $\mathbb{R}^3$, and let a be any real number. Then the following hold.

- (a) $(\mathbf{u} + \mathbf{v}) \times \mathbf{w} = \mathbf{u} \times \mathbf{w} + \mathbf{v} \times \mathbf{w}$
- (b) $\mathbf{u} \times (\mathbf{v} + \mathbf{w}) = \mathbf{u} \times \mathbf{v} + \mathbf{u} \times \mathbf{w}$
- (c) $\mathbf{u} \times \mathbf{v} = -\mathbf{v} \times \mathbf{u}$
- (d) $(a\mathbf{u}) \times \mathbf{v} = \mathbf{u} \times (a\mathbf{v}) = a(\mathbf{u} \times \mathbf{v})$
- (e) $\mathbf{u} \times \mathbf{v}$ is perpendicular to both $\mathbf{u}$ and $\mathbf{v}$
- (f) $\|\mathbf{u} \times \mathbf{v}\| = \|\mathbf{u}\|\|\mathbf{v}\|\sin\theta$ where θ is the angle between $\mathbf{u}$ and $\mathbf{v}$

Sometimes (e) and (f) can be taken as the geometric definition of the **cross product**. When $\mathbf{u}$ and $\mathbf{v}$ are not parallel, there is a unique straight line in $\mathbb{R}^3$ which is perpendicular to both $\mathbf{u}$ and $\mathbf{v}$, and knowing how to find such a line is quite important in studying three-dimensional geometry. The length of $\mathbf{u} \times \mathbf{v}$ also has a geometric meaning. It is equal to the area of the parallelogram with $\mathbf{u}$ and $\mathbf{v}$ as two sides. Therefore, the area of $\triangle ABC$ is equal to $\frac{1}{2}\|\vec{AB} \times \vec{AC}\|$. This shows the **cross product** is closely related to the area of planar objects.

When $\mathbf{u}$ and $\mathbf{v}$ are not parallel, there are two **vectors** which are perpendicular to both of them and having length $\|\mathbf{u}\|\|\mathbf{v}\|\sin\theta$. The two **vectors** are negative of each other. If we use these as the geometric definition of $\mathbf{u} \times \mathbf{v}$, we also need to specify the direction. This is given by the right-hand rule, i.e. if we use the fingers of the right hand to curl from $\mathbf{u}$ towards $\mathbf{v}$, the thumb will point in the direction of $\mathbf{u} \times \mathbf{v}$. For example, we have $\mathbf{i} \times \mathbf{j} = \mathbf{k}$, $\mathbf{j} \times \mathbf{k} = \mathbf{i}$ and $\mathbf{k} \times \mathbf{j} = -\mathbf{i}$.

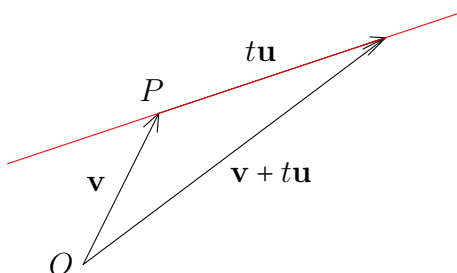
Note that (c) is different from the usual commutative property of multiplication. It is called the anticommutative property. In view of this, it is important that we do not mix up the order when writing a **cross product**.

1.2 Geometric Objects in the Three-dimensional Space

There are many geometric objects which only occur in $\mathbb{R}^3$. Even for those familiar objects like straight lines, they may have some different properties compared to those in $\mathbb{R}^2$. In this subsection, we shall go through everything that we need in this note, possibly with some additional things that we do not need.

Straight lines

A straight line can be defined by specifying a direction and a point on the straight line. Let P be a point in $\mathbb{R}^3$ with **position vector** $\mathbf{v}$, and let $\mathbf{u}$ be any nonzero **vector**. Then the equation of the line passing through P and parallel to $\mathbf{u}$ is $\mathbf{v} + t\mathbf{u}$, where t is a real parameter. This means a point lies on the straight line iff its **position vector** has the form $\mathbf{v} + t\mathbf{u}$ for some $t \in \mathbb{R}$. In other words, only the tip of the **vector** $\mathbf{v} + t\mathbf{u}$ lies on the straight line, but in general the **vector** $\mathbf{v} + t\mathbf{u}$ itself does not lie on the straight line.



As in $\mathbb{R}^2$, given any two distinct points in $\mathbb{R}^3$, there exists a unique straight line passing through both points. Indeed, if $\mathbf{u}$ and $\mathbf{v}$ are the **position vectors** of these two points, then $\mathbf{v} - \mathbf{u}$ is parallel to the straight line, and hence its equation is given by $\mathbf{u} + t(\mathbf{v} - \mathbf{u})$.

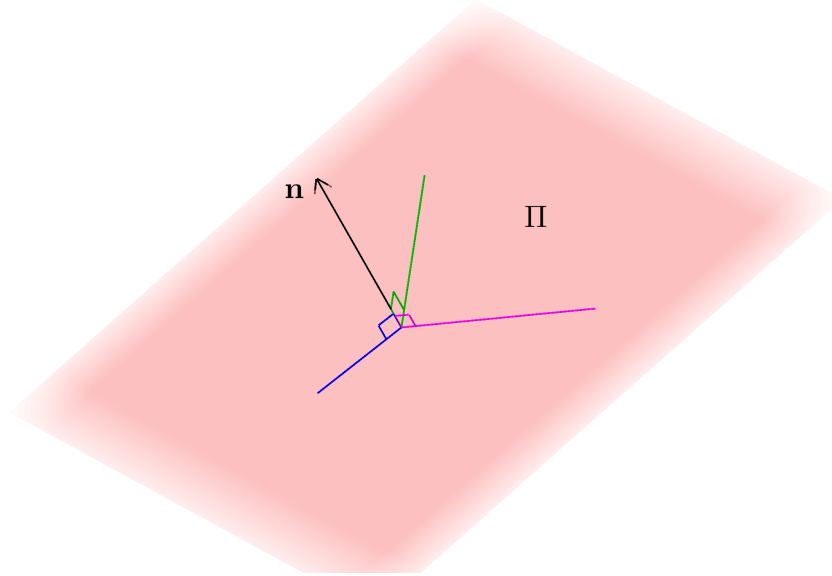
Given two straight lines with equations $\mathbf{u}_1 + t\mathbf{v}_1$ and $\mathbf{u}_2 + t\mathbf{v}_2$ (note that the two t 's are different parameters), since $\mathbf{v}_1$ and $\mathbf{v}_2$ represent their directions, we can talk about the angle between these lines in terms of the angle between $\mathbf{v}_1$ and $\mathbf{v}_2$. For example, these two lines are parallel iff $\mathbf{v}_1 // \mathbf{v}_2$, i.e. when $\mathbf{v}_1$ is a multiple of $\mathbf{v}_2$. Also, these two lines are perpendicular iff $\mathbf{v}_1 \perp \mathbf{v}_2$, i.e. $\mathbf{v}_1 \cdot \mathbf{v}_2 = 0$. Unlike the case in $\mathbb{R}^2$, where any two distinct lines are parallel or intersect, there are some pairs of straight lines in $\mathbb{R}^3$ which are non-parallel but do not intersect. These are called *skew lines* (歪斜線). For example, the straight lines $(0, 0, 0) + t(1, 0, 0)$ and $(0, 0, 1) + t(0, 1, 0)$ are non-parallel and do not intersect.

Planes

One important ingredient in $\mathbb{R}^3$ is a plane. This means something which is the same as $\mathbb{R}^2$ but possibly located at a different position. The usual $\mathbb{R}^2$ plane can be identified as the xy -plane $\{(x, y, 0) : x, y \in \mathbb{R}\}$, but in general there are infinitely many planes in $\mathbb{R}^3$. There are two common ways to understand a plane.

Let $\mathbf{u}$ and $\mathbf{v}$ be two **vectors** which are not parallel. Then the collection of all points with **position vector** $s\mathbf{u} + t\mathbf{v}$ where $s, t \in \mathbb{R}$ is a plane passing through the origin. In general, by using a translation, the equation of a plane (not necessarily passing through the origin) is given by $\mathbf{w} + s\mathbf{u} + t\mathbf{v}$. This can be seen as a simple generalization of straight lines, and can be further generalized as the definition of hyperplanes in $\mathbb{R}^n$.

However, the above algebraic meaning of planes is not useful in most situation. Rather than that, we can also fix a plane by choosing a line which is perpendicular to the plane and a point on the plane. Note that there is only one direction (or its negative) which is perpendicular to a given plane. Here, the perpendicularity means that the direction is perpendicular to any **vectors** lying entirely on the plane, not to the **vectors** whose tip lies on the plane. A **vector** perpendicular to a plane Π is called a *normal vector* or simply *normal*(法線) to Π .



Let $\mathbf{n} = (A, B, C)$ be a **normal** to a plane Π , and let (a, b, c) be a point on Π . A point (x, y, z) lies on Π iff $(x, y, z) - (a, b, c)$ is perpendicular to $\mathbf{n}$. Therefore, the equation of Π is

$$(A, B, C) \cdot (x - a, y - b, z - c) = 0.$$

This can be transformed to a standard form $Ax + By + Cz = D$. From this equation of the plane, we can easily find out a **normal** to it, which somehow specifies the direction of the plane. So this form is more useful in practice.

Suppose $\mathbf{u}$ and $\mathbf{v}$ are two non-parallel **vectors** lying entirely on a plane Π . Then since $\mathbf{u} \times \mathbf{v}$ is perpendicular to both $\mathbf{u}$ and $\mathbf{v}$, it is a **normal** to Π . The entries of $\mathbf{u} \times \mathbf{v}$ can be used to form the coefficients of the equation of Π . To determine the constant term, we only need to know one point lying on Π .

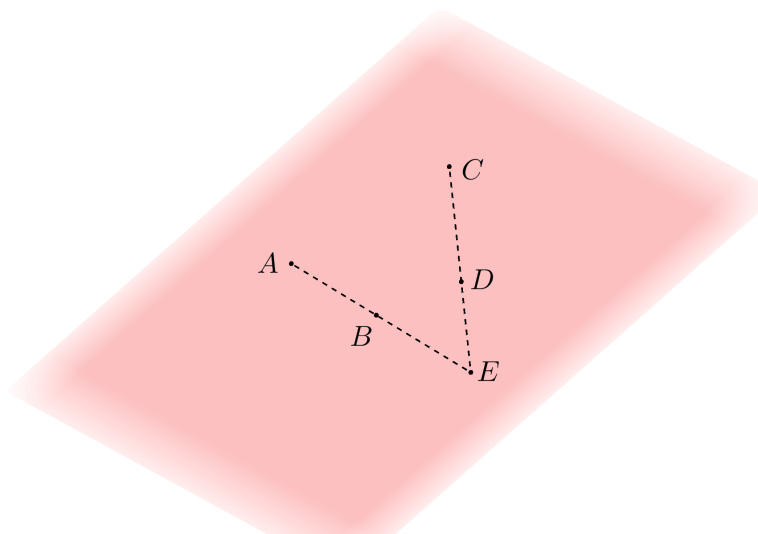
Given three non-collinear points X, Y, Z in $\mathbb{R}^3$, the **vectors** $\overrightarrow{XY}$ and $\overrightarrow{XZ}$ are non-parallel. Therefore, there exists a unique plane containing these **vectors**, i.e. there exists a unique plane containing X, Y, Z . It follows that any four distinct points need not lie on the same plane, and this leads to the following definition.

Definition 1.5. Let A, B, C, D be distinct points in $\mathbb{R}^3$. We say that they are *coplanar*(共面) if A, B, C, D lie on the same plane.

In many problems, it is useful to show that some points lie on the same plane, since afterwards we can reduce the problem to a usual geometry problem on the plane. A common way to prove that four points are **coplanar** is given below.

Proposition 1.4. Let A, B, C, D be distinct points in $\mathbb{R}^3$. They are **coplanar** iff $AB \parallel CD$ or AB intersects CD .

Equivalently, this means AB and CD is not a pair of **skew lines**. Note that we can permute the points A, B, C, D in any ways to apply this result. This means we can consider the pairs (AC, BD) or (AD, BC) instead.



Proof. If $AB \parallel CD$, let Π be the plane containing A, B, C . Then the **normal** to Π is perpendicular to AB , hence to CD , and so D also lies on Π .

If AB intersects CD at E , let Π be the plane containing A, B, C . Then Π contains the line AB , hence it contains E , and also the line CE . Thus, Π contains D .

If A, B, C, D are **coplanar**, the result is obvious from our knowledge in plane geometry. $\square$

Next, given a line and a point X not on the line, there exists a unique plane containing them. This is because we can find any two points Y, Z on the line, so that X, Y, Z determine a plane which clearly contains the line YZ . The uniqueness also follows easily.

In addition, two planes are said to be parallel if they have parallel **normals**. Equivalently, the two planes have equations $Ax + By + Cz = D_1$ and $Ax + By + Cz = D_2$ respectively. We can also say that a line is parallel to a plane, if the line is parallel to some line lying on the plane. A plane has no intersection with a parallel plane or a parallel line.

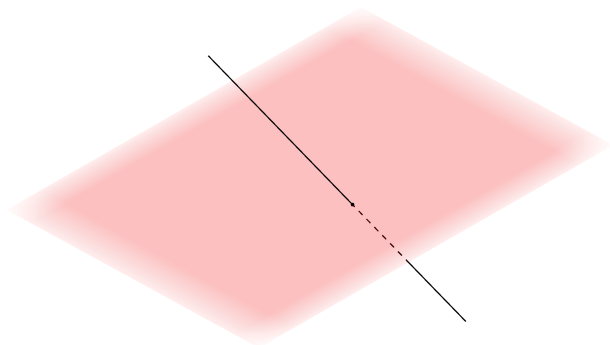
The following lists out some facts about lines and planes, which may be needed when we solve some three-dimensional geometric problems.

Proposition 1.5. Let P be a point, let ℓ, ℓ_1, ℓ_2 be straight lines, and let Π, Π_1, Π_2 be planes in $\mathbb{R}^3$. Then the following hold.

- (a) If ℓ and Π are non-parallel, then their intersection is a point.
- (b) If Π_1 and Π_2 are non-parallel, then their intersection is a straight line.
- (c) For any ℓ and Π , there exists a plane containing ℓ which is perpendicular to Π .
- (d) If ℓ_1 and ℓ_2 are **skew lines**, and P does not lie on ℓ_1 or ℓ_2 , then there exists a line passing through P lying on the same plane as ℓ_1 , and also lying on the same plane as ℓ_2 .
- (e) If ℓ_1 and ℓ_2 are **skew lines**, then there exists a pair of parallel planes passing through ℓ_1 and ℓ_2 respectively.
- (f) If ℓ_1 and ℓ_2 are **skew lines**, then there exists a straight line perpendicular to both of them and intersecting each of them.

Proof.

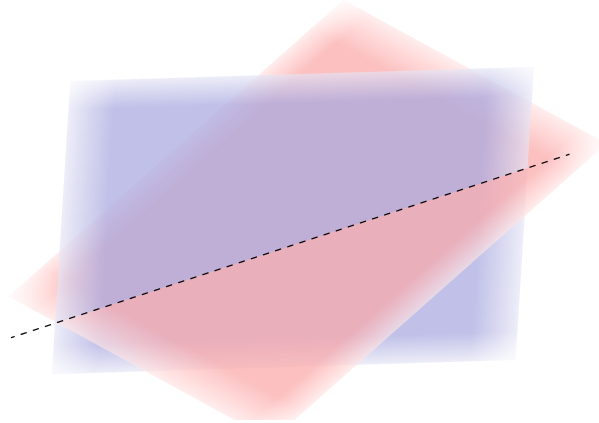
- (a) Let the equation of ℓ be $\mathbf{x} = \mathbf{u} + t\mathbf{v}$, and that of Π be $\mathbf{n} \cdot \mathbf{x} = c$. The intersection corresponds to the solution of $\mathbf{n} \cdot (\mathbf{u} + t\mathbf{v}) = c$, i.e. $t = \frac{c - \mathbf{n} \cdot \mathbf{u}}{\mathbf{n} \cdot \mathbf{v}}$ (by assumption, $\mathbf{n} \cdot \mathbf{v} \neq 0$). There is a unique solution, which means the intersection is a single point.



- (b) This basically means if we solve the system

$$\begin{cases} A_1x + B_1y + C_1z = D_1, \\ A_2x + B_2y + C_2z = D_2, \end{cases}$$

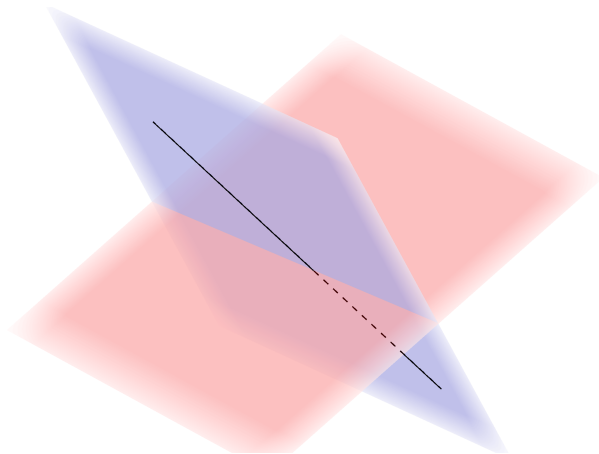
where $\mathbf{n}_1 = (A_1, B_1, C_1)$ and $\mathbf{n}_2 = (A_2, B_2, C_2)$ are not parallel, the general solution has the form $(x, y, z) = (x_0, y_0, z_0) + t(x', y', z')$ for some constant **vectors** (x_0, y_0, z_0) and $(x', y', z') \neq \mathbf{0}$. The details are omitted. Note that the line of intersection must be perpendicular to both $\mathbf{n}_1$ and $\mathbf{n}_2$, and hence it is parallel to $\mathbf{n}_1 \times \mathbf{n}_2$.



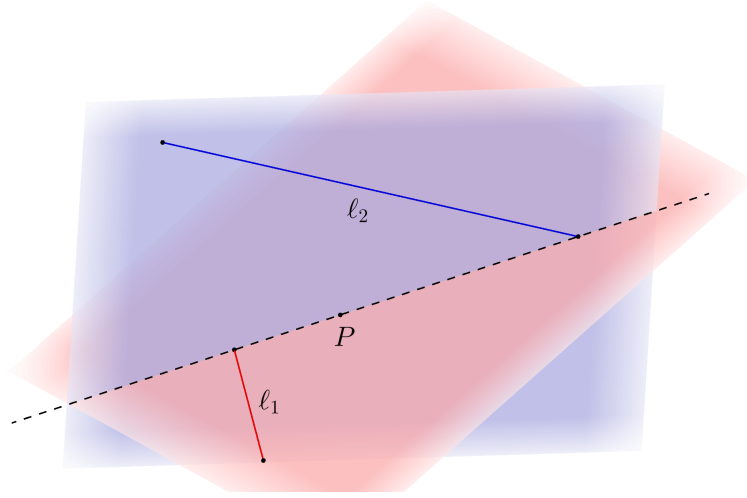
- (c) The angle between two planes is defined in a moment. Two planes are parallel iff their **normals** are perpendicular. Let $\mathbf{v}$ be a **vector** parallel to ℓ , and let $\mathbf{n}$ be a **normal** to Π .

If $\mathbf{v} \parallel \mathbf{n}$, we can choose any plane containing ℓ , so that its **normal** is perpendicular to ℓ and hence to $\mathbf{n}$.

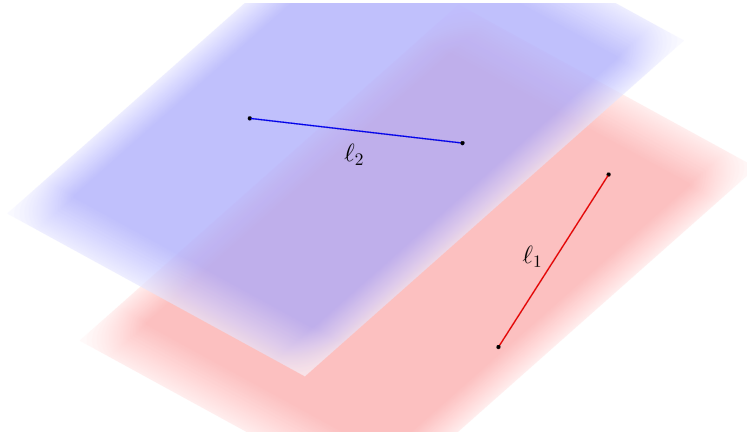
If $\mathbf{v}$ and $\mathbf{n}$ are not parallel, the **vector** $\mathbf{v} \times \mathbf{n}$ is perpendicular to both $\mathbf{v}$ and $\mathbf{n}$, so the plane with **normal** $\mathbf{v} \times \mathbf{n}$ and passing through any point on ℓ must contain ℓ and is perpendicular to Π . In this case, the required plane is also uniquely determined.



- (d) Let Π_1 be the plane containing P and ℓ_1 . Let Π_2 be the plane containing P and ℓ_2 . These planes are distinct since ℓ_1 and ℓ_2 are **skew**. As P lies on both planes, the planes intersect at a line ℓ . Thus, ℓ lies on the same plane Π_1 as ℓ_1 , and lies on the same plane Π_2 as ℓ_2 . Note that in the case when ℓ is not parallel to ℓ_1 and ℓ_2 , this gives a line passing through P which intersects ℓ_1 and intersects ℓ_2 . In view of this construction, the line ℓ is also unique.



- (e) Let $\mathbf{u}$ and $\mathbf{v}$ be **vectors** parallel to ℓ_1 and ℓ_2 respectively. Then $\mathbf{n} = \mathbf{u} \times \mathbf{v}$ is perpendicular to both ℓ_1 and ℓ_2 . So there exists a plane with **normal** $\mathbf{n}$ containing ℓ_1 , and there exists a plane with **normal** $\mathbf{n}$ containing ℓ_2 . These planes are parallel to each other. In view of this construction, the planes are uniquely determined.

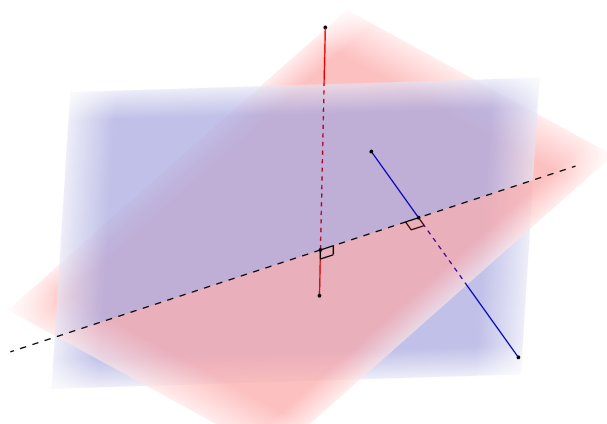


- (f) We draw any other line ℓ_3 which is perpendicular to both ℓ_1 and ℓ_2 , and not intersecting any of them. There are infinitely many such lines but its direction is fixed by using **cross product**. Note that the required line is also parallel to ℓ_3 .

Using the construction in (e), we can find a plane Π_1 containing ℓ_1 and parallel to ℓ_3 , and a plane Π_2 containing ℓ_2 and parallel to ℓ_3 . The planes Π_1 and Π_2 are non-parallel since ℓ_1, ℓ_2, ℓ_3 are pairwise **skew**. The intersection line of Π_1 and Π_2 is the required line because it must be parallel to ℓ_3 (if not, Π_1 and Π_2 contain two pairs of parallel lines in different directions, which implies they are parallel planes), and it must intersect ℓ_1 and ℓ_2 since it lies on a plane with one of them.

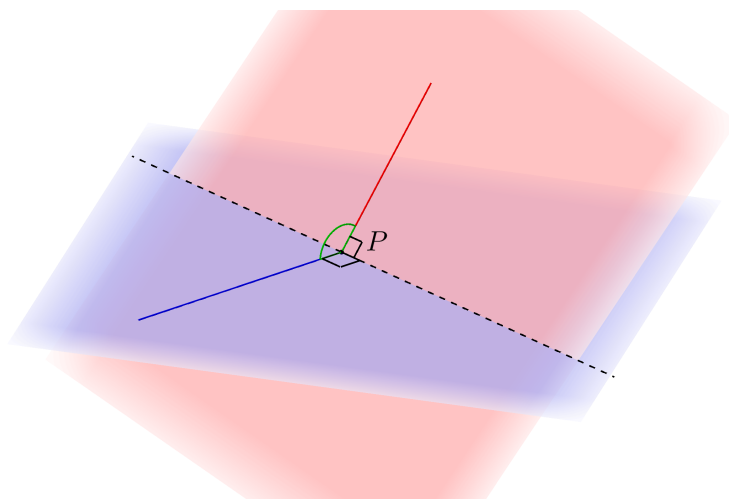
It can be shown that the minimum distance between a pair of **skew lines** exists, which is defined as the minimum distance of PQ with P on ℓ_1 and Q on ℓ_2 .

By considering some feet of perpendiculars, it is obvious that such a minimum distance can only be attained when PQ is perpendicular to both ℓ_1 and ℓ_2 . So the construction in this part tells us how to find the line of minimum distance between ℓ_1 and ℓ_2 . The uniqueness of this line follows from this geometric interpretation.



□

Definition 1.6. Let Π_1 and Π_2 be two planes intersecting at a line ℓ . Let P be any point on ℓ . Construct a line ℓ_1 on Π_1 which is perpendicular to ℓ at P . Construct a line ℓ_2 on Π_2 which is perpendicular to ℓ at P . The angle between Π_1 and Π_2 , called the *dihedral angle* (二面角) between them, is defined as the angle between ℓ_1 and ℓ_2 .



There are in fact two angles between a pair of lines. Some may restrict the angle between lines or planes to be the angle $\leq 90^\circ$. But it is still fine to call both θ and $180^\circ - \theta$ the angle between the lines or planes.

Note that the **dihedral angle** between two planes is equal to the angle between the **normals** to these planes. This can be seen by rotating the red line and blue line in

the above figure by 90° about the axis ℓ . The lines will become the **normals** to the planes passing through P . This gives a simple way to compute the **dihedral angle**.

Problem 1.3. Find the acute **dihedral angle** between the planes $x + 2y + 2z = 3$ and $2x - 3y - 6z = 5$.

Analysis. As mentioned, it suffices to find the angle between the **normals** to the planes. So we need to use the **dot product**.

Solution. The **normals** to the planes $x + 2y + 2z = 3$ and $2x - 3y - 6z = 5$ are $\mathbf{u} = (1, 2, 2)$ and $\mathbf{v} = (2, -3, -6)$ respectively. Let θ be the angle between these **vectors**. Then

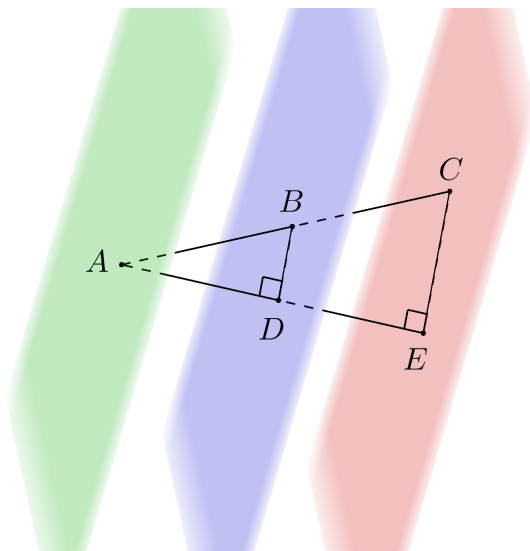
$$\cos \theta = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\| \|\mathbf{v}\|} = \frac{-16}{(3)(7)} = -\frac{16}{21}.$$

So the angle θ is obtuse, and the acute angle between the planes is $\pi - \theta = \cos^{-1} \left(\frac{16}{21} \right)$.

Problem 1.4. Let ℓ_1 and ℓ_2 be a pair of **skew lines**. Let Π_1, Π_2, Π_3 be parallel planes cutting ℓ_1 at A, B, C and cutting ℓ_2 at X, Y, Z respectively. Prove that $\frac{AB}{BC} = \frac{XY}{YZ}$.

Analysis. This is the analogue of the intercept theorem in $\mathbb{R}^2$, and can also be proved using similar triangles. Note that it suffices to show $\frac{AB}{BC}$ is a constant.

Solution.



Construct a line passing through A and perpendicular to the three planes. Suppose this line meets Π_2 and Π_3 at D and E respectively. Note that A, B, C, D, E are **coplanar**, and $\angle BDA = \angle CEA = 90^\circ$. Thus, $\triangle ABD \sim \triangle ACE$, and $\frac{AB}{BC} = \frac{AD}{DE}$.

This is the ratio of the distance between Π_1, Π_2 to the distance between Π_2, Π_3 , which is a constant. Therefore, we must have $\frac{AB}{BC} = \frac{XY}{YZ}$.

Remark. This example tells us the first idea of solving problems in $\mathbb{R}^3$, i.e. consider a plane passing through some of the points and work on the plane geometry.

Problem 1.5. (Carnot's theorem(卡诺定理)) Let A, B, C, D be four non-coplanar points in $\mathbb{R}^3$. Let W, X, Y, Z be points on AB, BC, CD, DA respectively. Prove that W, X, Y, Z are coplanar iff

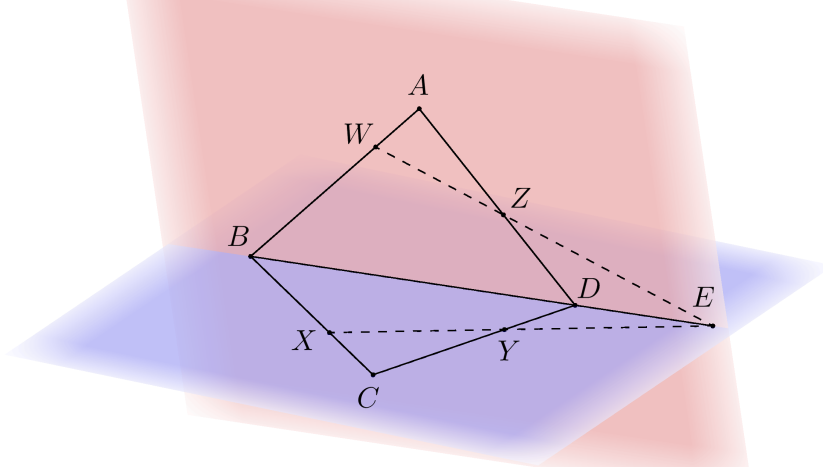
$$\frac{AW}{WB} \times \frac{BX}{XC} \times \frac{CY}{YD} \times \frac{DZ}{ZA} = 1, \quad (1)$$

where the lengths are expressed in directed length.

Analysis. This looks like Menelaus' theorem in plane geometry, so we can think of using Menelaus' theorem by drawing some triangles.

Solution. Firstly, suppose W, X, Y, Z are coplanar.

- If the plane $WXYZ$ is parallel to BD , then there exist a pair of parallel planes, one passing through WZ and the other passing through BD . As a special case of problem 1.4, we have $\frac{AW}{WB} = \frac{AZ}{ZD}$. Similarly, we have $\frac{CX}{XB} = \frac{CY}{YD}$. These readily imply (1).



- If the plane $WXYZ$ is not parallel to BD , let E be the intersection of this plane and the line BD . Then E lies on WZ and XY because WZ is the intersection of the plane $WXYZ$ and the plane ABD , etc. Applying Menelaus' theorem to $\triangle ABD$ and $\triangle CBD$ respectively, we obtain

$$\begin{aligned} \frac{AW}{WB} \times \frac{BE}{ED} \times \frac{DZ}{ZA} &= 1, \\ \frac{CX}{XB} \times \frac{BE}{ED} \times \frac{DY}{YC} &= 1. \end{aligned}$$

The quotient of these imply (1).

Secondly, suppose (1) holds.

- If $\frac{AW}{WB} = \frac{AZ}{ZD}$, then $\frac{CX}{XB} = \frac{CY}{YD}$. These imply $WZ \parallel BD \parallel XY$, and so W, X, Y, Z are **coplanar**.
- If $\frac{AW}{WB} \neq \frac{AZ}{ZD}$, then $\frac{CX}{XB} \neq \frac{CY}{YD}$. Let WZ meet BD at E_1 , and let XY meet BD at E_2 . Applying Menelaus' theorem to $\triangle ABD$ and $\triangle CBD$ respectively, we easily obtain $\frac{BE_1}{E_1D} = \frac{BE_2}{E_2D}$ in directed length. Therefore, $E_1 = E_2$. This means WZ intersects XY . By proposition 1.4, W, X, Y, Z are **coplanar**.

Polyhedrons

The analogue of polygons in $\mathbb{R}^3$ is *polyhedrons*(多面體). A **polyhedron** is a solid in the three-dimensional space whose surface is composed of a finite number of polygons. Each such polygon is called a *face*(面) of the **polyhedron**. Each side of a face is called an *edge*(稜), and each vertex of a face is still called a *vertex*(頂點).

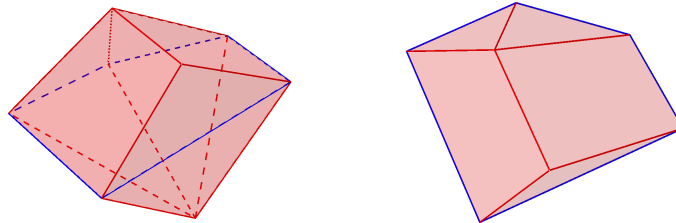
We mainly talk about *convex polyhedrons*(凸多面體) in this note. This means a **polyhedron** such that the segment joining any two points on or inside this **polyhedron** lies within the **polyhedron**. Equivalently, this means it is the convex hull of a finite number of points in $\mathbb{R}^3$ which do not lie on the same plane.

The following gives a well-known result relating the number of vertices, the number of edges and the number of faces of a **convex polyhedron**.

Theorem 1.2. (Euler's polyhedron formula(歐拉公式)) Let V, E, F be the number of vertices, the number of edges and the number of faces of a **convex polyhedron** respectively. Then

$$V - E + F = 2.$$

Proof. Take any closed path C formed by some of the edges of the **convex polyhedron** $\mathcal{P}$ that has no self-intersection. For example, start from any vertex and travel along distinct edges arbitrarily, until we pass through the same vertex twice. Then we can take the path with this vertex as initial and terminal points. Let C contain n vertices.



The closed path C naturally divides $\mathcal{P}$ into two halves. We focus on one half of $\mathcal{P}$. Let V_1, E_1, F_1 be the number of vertices, the number of edges and the number of faces on this half excluding those on C . Note that we need to prove $V_1 - E_1 + F_1 = 1$, since if this holds, we have the same relation $V_2 - E_2 + F_2 = 1$ for the other half, and so

$$\begin{aligned} V - E + F &= (V_1 + V_2 + n) - (E_1 + E_2 + n) + (F_1 + F_2) \\ &= (V_1 - E_1 + F_1) + (V_2 - E_2 + F_2) = 2. \end{aligned}$$

Note that we can deform the solid $\mathcal{P}$ in a continuous way, as long as the vertices, edges and faces keep the same relations. Therefore, we may assume C bounds a polygon on a plane, and assume this path is sufficiently large. Then the projection of every other vertex on this plane lies inside C . We can visualize one half of $\mathcal{P}$ using the aerial view. This means we can regard this half of $\mathcal{P}$ as a planar figure.

The V_1 vertices together with their edges partition the polygon bounded by C into some polygons. Suppose these polygons have $a_1, a_2, \dots, a_{F_1}$ sides respectively. By considering the sum of internal angles, a simple double counting yields

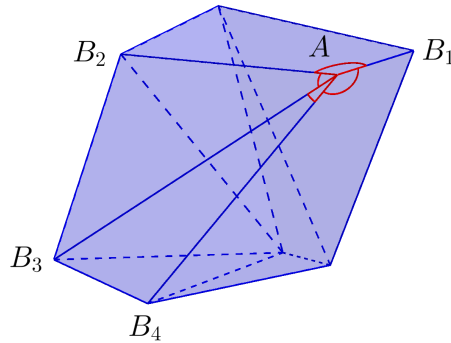
$$\sum_{j=1}^{F_1} 180(a_j - 2) = 180(n - 2) + 360V_1.$$

This implies $a_1 + a_2 + \dots + a_{F_1} - 2F_1 = n - 2 + 2V_1$. Now, each interior edge belongs to exactly two polygons, while each edge of C belongs to only one polygon. Therefore, we have $E_1 = \frac{1}{2}(a_1 + a_2 + \dots + a_{F_1} - n)$. We finally obtain

$$V_1 - E_1 + F_1 = \frac{1}{2}(2V_1 - a_1 - a_2 - \dots - a_{F_1} + n + 2F_1) = 1.$$

□

Next, we look for some results about the interior angles of a **polyhedron**.



Proposition 1.6. Let A be a vertex of a **convex polyhedron**. Let $B_1, B_2, \dots, B_n$ be the vertices adjacent to A in this order. Then we have the relation

$$\angle B_1AB_2 + \angle B_2AB_3 + \dots + \angle B_{n-1}AB_n + \angle B_nAB_1 < 360^\circ.$$

This seems to be obvious intuitively, because the angle sum at a point on a plane is 360° . But a rigorous proof is rather tedious. We will leave the proof to the end of this subsection.

Theorem 1.3. (Descartes' theorem(笛卡兒定理)) Given a **convex polyhedron**, the *angular defect* at a vertex is defined as 360° minus the sum of angles at this vertex. The total **angular defect** of a **convex polyhedron** must be 720° .

Proof. Let the F faces have $a_1, a_2, \dots, a_F$ sides respectively. Then the number E of edges of the **convex polyhedron** $\mathcal{P}$ is $\frac{1}{2}(a_1 + a_2 + \dots + a_F)$. Let the angles at a vertex V_i be $\theta_{i1}, \theta_{i2}, \dots, \theta_{is_i}$ respectively. By definition, the total **angular defect** of $\mathcal{P}$ is

$$\sum_{i=1}^V (360^\circ - \theta_{i1} - \theta_{i2} - \dots - \theta_{is_i}) = 360^\circ V - \sum_{i=1}^V \sum_{j=1}^{s_i} \theta_{ij}.$$

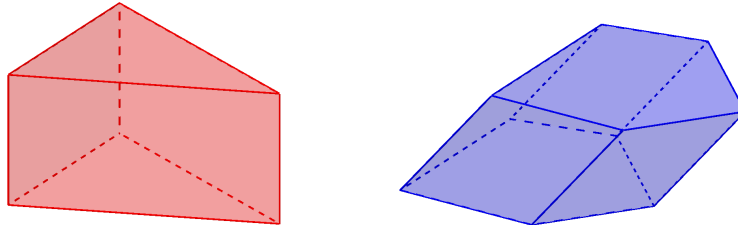
The double sum is the sum of all angles of the faces. Hence, the total **angular defect** is

$$\begin{aligned} 360^\circ V - \sum_{i=1}^F (a_i - 2)180^\circ &= 360^\circ V - (a_1 + a_2 + \dots + a_F - 2F)180^\circ \\ &= 360^\circ V - (2E - 2F)180^\circ \\ &= (V - E + F)360^\circ = 720^\circ \end{aligned}$$

by **Euler's polyhedron formula**. □

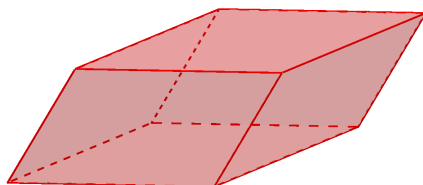
Two kinds of commonly seen **convex polyhedrons** are **prism** and **pyramid**.

A *prism*(角柱) is composed of two identical polygonal bases which are translation of each other located at different planes, together with other faces formed by joining the corresponding vertices of the bases. Note that each face except the bases must be a parallelogram. Some definitions require those faces to be rectangles. The volume of a **prism** is the base area times the height (which is the distance between the two bases). **Prisms** are usually named according to their bases. For example, the figures below show a triangular **prism** and a pentagonal **prism**.

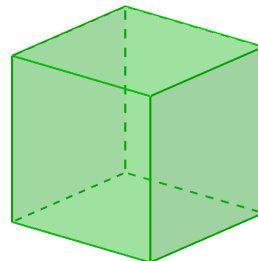
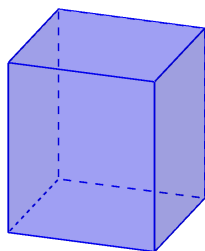
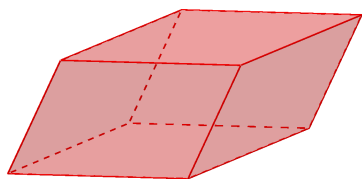


Here are some **prisms** which are worth to mention. A *parallelepiped*(平行六面體) is a **prism** whose bases are parallelograms. All its 6 faces are parallelograms, and

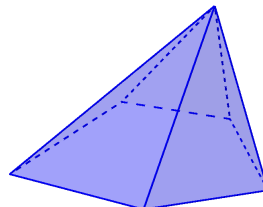
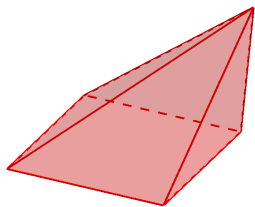
the 3 pairs of opposite faces are parallel and identical to each other. The 12 edges can be partitioned into 3 groups of parallel edges with equal length. Analogous to a parallelogram, the midpoints of all 4 pairs of opposite vertices coincide.



In particular, a *rhombohedron*(菱面體) is a **parallelepiped** whose 6 faces are all rhombuses. A *cuboid*(長方體) is a **parallelepiped** whose 6 faces are all rectangles. A *cube*(立方體) is a **parallelepiped** whose 6 faces are all squares. So a **cube** is both a **rhombohedron** and a **cuboid**.



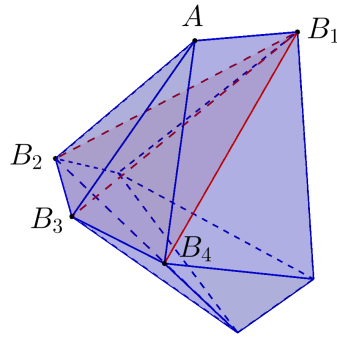
Next, a *pyramid*(角錐) is formed by connecting the vertices of a polygonal base with a point not lying on the same plane as the base. Note that each face except the base must be a triangle. A *right pyramid* is a **pyramid** whose top vertex lies exactly above the centroid of the base. The area of a **pyramid** is $\frac{1}{3}$ times the base area times the height (which is the distance from the top vertex to the plane containing the base). Again, **pyramids** are usually named according to their bases. For example, the figures below show a square **pyramid** and a pentagonal **pyramid**.



A triangular **pyramid** is more commonly known as a *tetrahedron*(四面體). Each of its 4 faces is a triangle, and it is the **polyhedron** with the smallest number of edges. It is the analogue of a triangle in $\mathbb{R}^3$. For example, just like every convex polygon in $\mathbb{R}^2$ can be decomposed into some triangles (with vertices being the vertices of the polygon), every **convex polyhedron** can be decomposed into some **tetrahedrons**. Here

illustrates the simple proof of this fact.

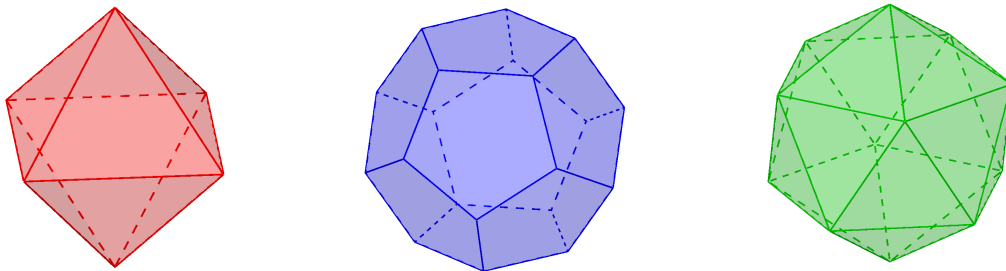
We pick a vertex A of the **convex polyhedron**, and label its adjacent vertices as $B_1, B_2, \dots, B_n$ in order, where $n \geq 3$. Then we remove the **tetrahedrons** $AB_1B_kB_{k+1}$ for $k = 2, 3, \dots, n-1$. After doing this, the remaining **polyhedron** is still **convex**, and it has one vertex less (vertex A) than the original **polyhedron**. Repeating the same process, we eventually obtain a **convex polyhedron** with only 4 vertices, which is itself a **tetrahedron**. The dissection is complete.



More about **tetrahedrons** will be mentioned in section 2.

Lastly, a *regular polyhedron*(正多面體) is a **polyhedron** whose every face is the same regular polygon, and whose vertex is the common vertex of the same number of faces arranged in the same way. All its edges must have the same length. Thus, this is a highly symmetric solid. A **convex regular polyhedron** is also known as a *Platonic solid*(柏拉圖立體).

A **regular tetrahedron** and a **cube** are both **Platonic solids**. The following figures show 3 other **Platonic solids**, called a *regular octahedron*(正八面體), a *regular dodecahedron*(正十二面體), and a *regular icosahedron*(正二十面體) respectively.



Theorem 1.4. There are only 5 different **Platonic solids** up to the size, namely, **regular tetrahedrons**, **cubes**, **regular octahedrons**, **regular dodecahedrons** and **regular icosahedrons**.

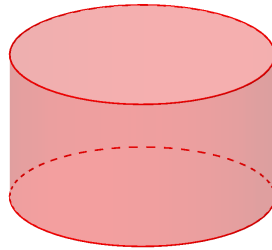
Proof. Suppose each vertex of a **Platonic solid** is the common vertex of k regular polygons of n sides. By proposition 1.6, the angle sum at a vertex is less than 360° .

This means $k \cdot \frac{(n-2)180^\circ}{n} < 360^\circ$, i.e. $k(n-2) < 2n$. Note that $k \geq 3$ as a vertex cannot be a common vertex of fewer than 3 faces. This gives the only solutions $(k, n) = (3, 3), (3, 4), (4, 3), (3, 5), (5, 3)$. For each solution, there is a unique way to combine the k regular polygons of n sides at a vertex, and so this gives rise to at most one **Platonic solid** up to the size. Indeed, these solutions correspond to the 5 **Platonic solids** as mentioned. So there are exactly 5 **Platonic solids**. $\square$

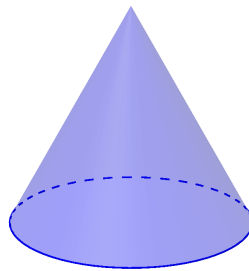
Cylinders, cones and spheres

Apart from polygons, the circles form another kind of important geometric object in $\mathbb{R}^2$. Analogously, we have some objects in $\mathbb{R}^3$ which are built up by circles.

A *cylinder*(圓柱) is a generalization of a **prism** with circles as bases. It is more common to talk about right **cylinders**, where the top base lies exactly above the bottom base. The volume of a **cylinder** is the base area times the height.

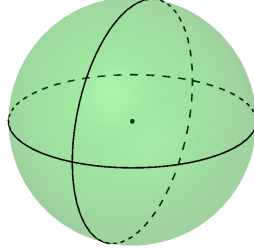


A *cone*(圓錐) is a generalization of a **pyramid** with circle as the base. It is more common to talk about right circular **cones**, where the top vertex lies exactly above the centre of the base circle. The volume of a **cone** is $\frac{1}{3}$ times the base area times the height.



A *sphere*(球面) is the set of all points in $\mathbb{R}^3$ which have the same distance from a fixed point. This is a simple generalization of circles, and similar definitions apply. The fixed point is called the *centre*(球心) of the **sphere**, while the distance R is called the *radius*(半徑). Any segment joining two points on the **sphere** has distance $\leq 2R$. Equality holds when the segment passes through the centre, in which case the

segment is called a *diameter*(直徑). Two points on a **sphere** are called *antipodes*(對蹠點) of each other if they are the endpoints of a diameter. Also, a *hemisphere*(半球) is half of a **sphere**.



Just like the unit circle in $\mathbb{R}^2$ with centre $(0, 0)$ has the parametrization $(\cos \theta, \sin \theta)$ using the polar coordinates, the unit **sphere** (a **sphere** with radius 1) in $\mathbb{R}^3$ with centre $(0, 0, 0)$ also has a parametrization $(\sin \varphi \cos \theta, \sin \varphi \sin \theta, \cos \varphi)$ using the spherical coordinates, where $0 \leq \varphi \leq \pi$ and $0 \leq \theta \leq 2\pi$. Using multivariable calculus, we can show that the volume of a **sphere** with radius R is $\frac{4}{3}\pi R^3$, and its surface area is $4\pi R^2$.

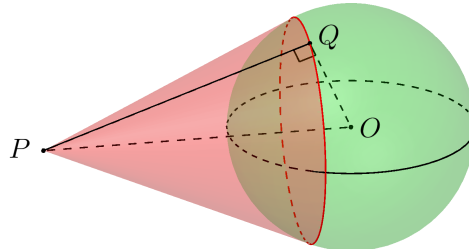
In subsection 2.3, we will show that any 4 non-**coplanar** points lie on a uniquely determined **sphere**. In general, we say that some points are *cospherical*(共球) if they lie on the same **sphere**.

The following lists out some simple facts about **spheres**.

Proposition 1.7. Let S be a **sphere**. Then the following hold.

- (a) Any straight line meets S at 0, 1 or 2 points.
- (b) Any plane meets S at 0, 1 or an infinite number of points. In the last case, the intersection must be a circle.
- (c) Any **sphere** meets S at 0, 1 or an infinite number of points. In the last case, the intersection must be a circle.

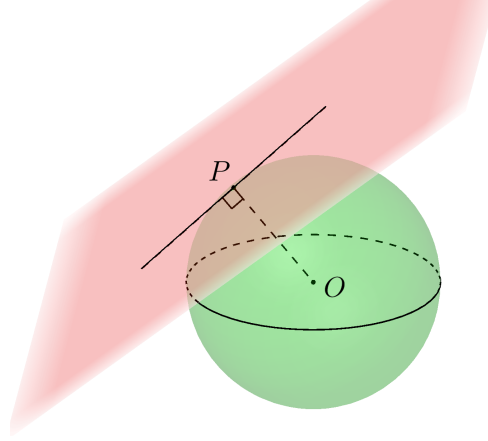
The results can be proved algebraically. The details are omitted.



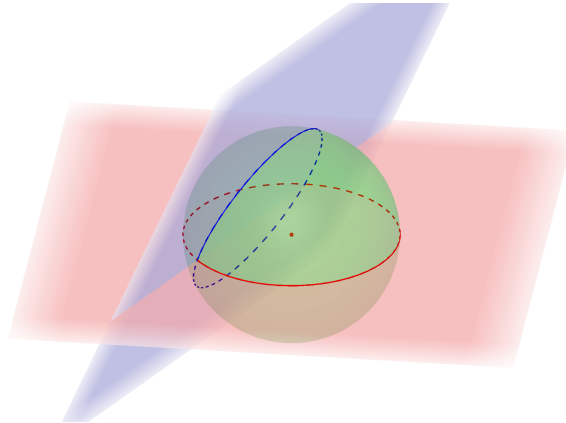
A line, a plane or a **sphere** meeting S at exactly 1 point is said to be *tangent*(相切) to S . Through any point P outside S , we can draw an infinite number of **tangent**

lines to S . These **tangent lines** form a **cone** with vertex P . Let O be the centre of S , and let Q be the contact point of a **tangent line** from P . As in the two-dimensional case, we have $OQ \perp PQ$, and hence $PQ = \sqrt{OP^2 - OQ^2}$ is a constant.

On the other hand, for any point P on S , there exists a unique **tangent plane** to S passing through P . The **normal** to this plane is parallel to OP . All **tangent lines** passing through P lie on this plane.



If a plane passes through the centre of S , its intersection with S is called a *great circle*(大圓) because it is the largest circle that we can draw on S . Any other circles on S (those formed by the intersection of S and a plane not passing through the centre of S) are called *small circles*(小圓).



Proposition 1.8. Let S be a **sphere**. Then the following hold.

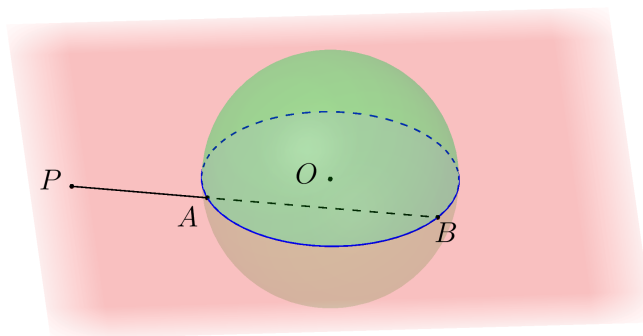
- (a) For any two points on S , there exists a **great circle** passing through both of them. If the two points are not **antipodes**(對蹠點) of each other, then the **great circle** is uniquely determined. Otherwise, there are infinitely many such **great circles**.
- (b) The shortest path between two points on S is the minor arc of a **great circle** passing through these points.

For (a), it suffices to note that we can and can only take a plane passing through the two points and the centre of S . Such a plane is unique unless these three points are collinear, i.e. when the two points are **antipodes**. A common proof of (b) requires calculus, and is omitted.

Problem 1.6. (Power of a point theorem(圓幂定理)) Let S be a **sphere** with centre O and radius R , and let P be a point outside S . A line passing through P meets S at two distinct points A and B . Prove that $PA \times PB = OP^2 - R^2$.

Analysis. We should be familiar with this result in $\mathbb{R}^2$. Instead of following the traditional proof, we show how to use the technique of **great circles**.

Solution.



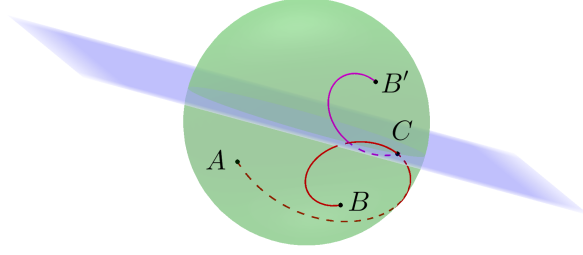
Consider a **great circle** Γ of S containing A and B . The **power** of P with respect to(w.r.t.) Γ is $PA \times PB$. As a **great circle**, the centre and radius of Γ are the same as those of S . Hence, by the power of a point theorem for circles, we have $PA \times PB = OP^2 - R^2$.

Remark. Similar results analogous to the $\mathbb{R}^2$ case hold when P lies on or inside S . It is therefore natural to define the *power*(圓幂) of P w.r.t. S as $OP^2 - R^2$.

Problem 1.7. (USAMO 1974 Problem 3 modified) Two points on a **sphere** S of radius 1 are joined by a curve on S having length less than π . Prove that the curve is contained on some **hemisphere** of S .

Analysis. As this problem is about an arbitrary curve on a **sphere**, this suggests using proposition 1.8(b). Note that the length of any curve joining two **antipodal** points of S is less than π . So we can make use of **antipodal** points.

Solution. Let O be the centre of S . Label the two endpoints A and B . Let B' be the **antipode** of A on S . Consider the **mediator** (perpendicular bisector plane) Π of BB' . This means B' is the image of B under the reflection in Π . Note that O lies on Π , and so Π cuts S into two **hemispheres**. We claim that the curve lies on the **hemisphere** containing A .



Suppose on the contrary that the curve does not lie entirely on the **hemisphere** containing A . Let C be the first intersection point of the curve (starting from A) and Π . We reflect the remaining part of the curve from C to B in the plane Π to obtain a curve from C to B' . Then the curve from A to C and then to B' has the same length as the original curve from A to B . So we have obtained a curve on S between A and B' which has length less than π . But then the shortest path between the two **antipodal** points is the minor arc of a **great circle** passing through A and B' by proposition 1.8(b), which is indeed a semicircle with radius 1. So the length of any curve joining A and B' is at least π . This is a contradiction, and we are done.

Problem 1.8. (Proposition 1.6) Let A be a vertex of a **convex polyhedron**. Let $B_1, B_2, \dots, B_n$ be the vertices adjacent to A in this order. Prove that

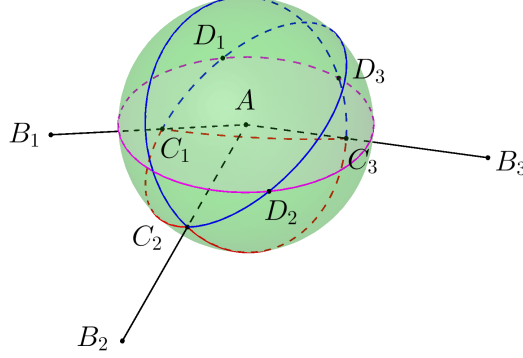
$$\angle B_1AB_2 + \angle B_2AB_3 + \dots + \angle B_{n-1}AB_n + \angle B_nAB_1 < 360^\circ.$$

Analysis. We return to the proof of this result. One may tend to use an intuitive ‘fact’: If we project A on a plane to obtain A' , then $\angle BAC < \angle BA'C$ for any B, C on the plane. This holds when $\triangle A'BC$ is acute, but is not true in general. Indeed, we will use some concepts about **spheres**.

Solution. Draw a sufficiently small **sphere** S with A as centre. Suppose S intersects the edges $AB_1, AB_2, \dots, AB_n$ at $C_1, C_2, \dots, C_n$ respectively. The intersection of S with the face having AB_1 and AB_2 as two sides is the arc of a **great circle** (the intersection of S and the plane AB_1B_2) joining C_1 and C_2 . Similarly, the intersection of S and the **convex polyhedron** $\mathcal{P}$ is a union of n arcs of some **great circles**. Note that the arcs must be minor arcs, and they form a closed curve on S (the red curves in the figure).

WLOG, let the radius of S be 1. Then the length of $\widehat{C_1C_2}$ is $2\pi \cdot \frac{\angle B_1AB_2}{360^\circ}$. Similar results hold for the length of the other arcs. Therefore, the problem is equivalent to proving the total length of $\widehat{C_1C_2}, \widehat{C_2C_3}, \dots, \widehat{C_{n-1}C_n}, \widehat{C_nC_1}$ is less than 2π .

Consider a **great circle** Γ of S such that $\mathcal{P}$ lies on one side of it. This is possible since $\mathcal{P}$ is **convex**. We now show that the circumference of Γ (which is 2π) is larger than the total length of the arcs. Let D_1 and D_2 be the intersection of Γ and the **great circle** containing $\widehat{C_1C_2}$. For $k = 3, 4, \dots, n$, let D_k be the intersection of Γ and



the **great circle** containing $\widehat{C_{k-1}C_k}$ such that $\widehat{C_{k-1}C_kD_k}$ is a minor arc. The point D_k is uniquely determined since $\widehat{C_{k-1}C_k}$ lies on one side of Γ . Note that $D_1, D_2, \dots, D_n$ must lie in this order on Γ .

We now use proposition 1.8(b) to deduce relations like $\widehat{C_2D_2} + \widehat{D_2D_3} > \widehat{C_2D_3}$, etc. We have

$$\begin{aligned}
2\pi &= \widehat{D_1C_1C_2D_2} + \widehat{D_2D_3 \cdots D_nD_1} \\
&= \widehat{D_1C_1C_2} + (\widehat{C_2D_2} + \widehat{D_2D_3}) + \widehat{D_3D_4 \cdots D_nD_1} \\
&> \widehat{D_1C_1C_2} + \widehat{C_2C_3D_3} + \widehat{D_3D_4 \cdots D_nD_1} \\
&= \widehat{D_1C_1C_2} + \widehat{C_2C_3} + (\widehat{C_3D_3} + \widehat{D_3D_4}) + \widehat{D_4D_5 \cdots D_nD_1} \\
&> \widehat{D_1C_1C_2} + \widehat{C_2C_3} + \widehat{C_3C_4D_4} + \widehat{D_4D_5 \cdots D_nD_1} \\
&> \dots \\
&> \widehat{D_1C_1C_2} + \widehat{C_2C_3} + \widehat{C_3C_4} + \dots + \widehat{C_{n-2}C_{n-1}} + \widehat{C_{n-1}C_nD_n} + \widehat{D_nD_1} \\
&= \widehat{C_1C_2} + \widehat{C_2C_3} + \widehat{C_3C_4} + \dots + \widehat{C_{n-2}C_{n-1}} + \widehat{C_{n-1}C_n} + (\widehat{C_nD_n} + \widehat{D_nD_1} + \widehat{D_1C_1}) \\
&> \widehat{C_1C_2} + \widehat{C_2C_3} + \widehat{C_3C_4} + \dots + \widehat{C_{n-2}C_{n-1}} + \widehat{C_{n-1}C_n} + \widehat{C_nC_1}.
\end{aligned}$$

This completes the proof.

2 Centres of Tetrahedrons

In this section, we explore more about the properties and some centres of **tetrahedrons**.

2.1 General Properties

Let $ABCD$ be a **tetrahedron**. Its volume $[ABCD]$ is equal to $\frac{1}{3}[BCD]h_A$, where h_A is the height from A to the base BCD . Another convenient way to express $[ABCD]$ is given by the formula

$$\frac{1}{6}|\vec{AB} \cdot (\vec{AC} \times \vec{AD})|.$$

In addition, analogous to Heron's formula, we can express $[ABCD]$ in terms of the side lengths of $ABCD$. Let $d_{AB} = AB$, etc. Then $[ABCD]$ can be expressed as the *Cayley-Menger determinant* (凱萊-門格行列式)

$$288[ABCD]^2 = \begin{vmatrix} 0 & 1 & 1 & 1 & 1 \\ 1 & 0 & d_{AB}^2 & d_{AC}^2 & d_{AD}^2 \\ 1 & d_{AB}^2 & 0 & d_{BC}^2 & d_{BD}^2 \\ 1 & d_{AC}^2 & d_{BC}^2 & 0 & d_{CD}^2 \\ 1 & d_{AD}^2 & d_{BD}^2 & d_{CD}^2 & 0 \end{vmatrix}.$$

This means $[ABCD]$ is the square root of a polynomial in $d_{AB}^2, d_{AC}^2, \dots, d_{CD}^2$.

Similar to the triangle inequality, we have the relation

$$[ABC] + [ABD] + [ACD] > [BCD].$$

In fact, by letting θ_{AB} be the **dihedral angle** between the faces ABC and ABD , etc., we have the identity

$$\begin{aligned} [BCD]^2 &= [ABC]^2 + [ABD]^2 + [ACD]^2 - 2[ABC][ABD]\cos\theta_{AB} \\ &\quad - 2[ABC][ACD]\cos\theta_{AC} - 2[ABD][ACD]\cos\theta_{AD}. \end{aligned}$$

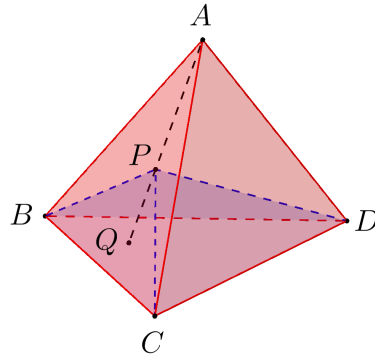
This is the cosine formula for **tetrahedrons**. We also have the sine law for **tetrahedrons**, e.g.

$$\sin \angle ABC \sin \angle ACD \sin \angle ADB = \sin \angle ABD \sin \angle ADC \sin \angle ACB.$$

These results are not useful for our purpose.

Proposition 2.1. Let $ABCD$ be a **tetrahedron**, and let Q be a point on the plane BCD . If P is a point on the line AQ , then

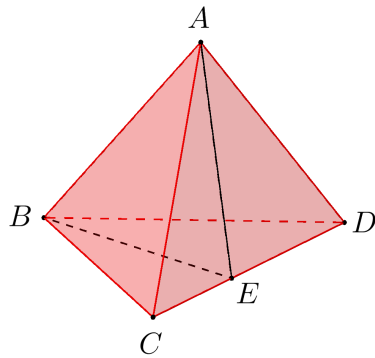
$$\frac{[PBCD]}{[ABCD]} = \frac{PQ}{AQ}.$$



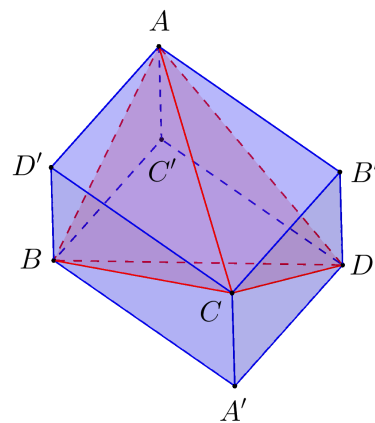
Proof. The two **tetrahedrons** $PBCD$ and $ABCD$ have a common base BCD . So their ratio of volumes is the ratio of the heights from P and A . By similar triangles, this is equal to $PQ : AQ$. $\square$

This extends the usual result for ratio of areas in $\mathbb{R}^2$. As a special case, if E is a point on the edge CD , then

$$\frac{[ABCE]}{[ABDE]} = \frac{CE}{DE}.$$



Next, we introduce an important solid arisen from a **tetrahedron**.



Definition 2.1. Let $ABCD$ be a **tetrahedron**. There exists a unique **parallelepiped** with A, B, C, D as four non-adjacent vertices. This is called the *circumscribed parallelepiped*(外接平行六面體) of $ABCD$.

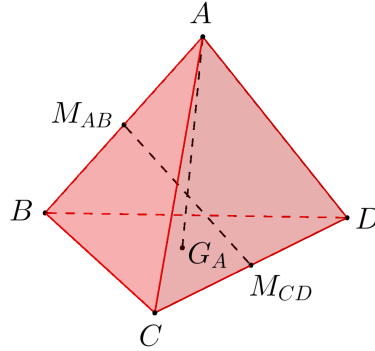
Label the other vertices of the **circumscribed parallelepiped** as A', B', C', D' such that $A, A'; B, B'; C, C'; D, D'$ are opposite vertices. The plane $AC'BD'$ is the unique plane passing through A, B which is parallel to CD . Similarly, the planes of all faces of this **circumscribed parallelepiped** are uniquely determined. The solid bounded by the six planes must be a **parallelepiped** because opposite planes are parallel. This gives the existence and uniqueness of the **circumscribed parallelepiped**.

In fact, we have already seen an application of **circumscribed parallelepipeds** in problem 1.1. In that example, $ABCD$ is a **regular tetrahedron**, and its **circumscribed parallelepiped** is a **cube**. Later we will see some more applications of this.

2.2 Centroid

We first define some kinds of medians, analogous to the $\mathbb{R}^2$ case.

Definition 2.2. Let $ABCD$ be a **tetrahedron**. Let G_A be the centroid of $\triangle BCD$. Then the segment AG_A is called a *median*(中線) of $ABCD$.



Definition 2.3. Let $ABCD$ be a **tetrahedron**. Let M_{AB} and M_{CD} be the midpoints of AB and CD respectively. Then the segment $M_{AB}M_{CD}$ is called a *bimedial*(對稜中線) of $ABCD$.

By definition, a **tetrahedron** has 4 **medians** and 3 **bimedians**.

Theorem 2.1. Let $ABCD$ be a **tetrahedron**. Then the following hold.

- (a) (Commandino's theorem(科曼迪諾定理)) The **medians** AG_A, BG_B, CG_C, DG_D of $ABCD$ are concurrent at a point G , and

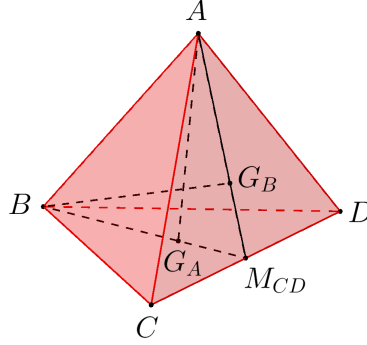
$$AG : GG_A = BG : GG_B = CG : GG_C = DG : GG_D = 3 : 1.$$

(b) The point G defined in (a) is the midpoint of each **bimedian** of $ABCD$.

As a result, all the **medians** and **bimedians** meet at a point G , called the *centroid*(重心) of $ABCD$. An algebraic proof to theorem 2.1 is very simple, because the **centroid** is the arithmetic mean of the 4 vertices. But we shall provide a geometric proof below.

Proof.

(a) Let M_{AB} be the midpoint of AB , etc. We first prove that AG_A and BG_B intersect. Indeed, since AG_B and BG_A are medians of $\triangle ACD$ and $\triangle BCD$ respectively, both of them pass through M_{CD} . By proposition 1.4, the points A, B, G_A, G_B are **coplanar**. Also, note that G_B and G_A cut the segments AM_{CD} and BM_{CD} in the ratio $2 : 1$. So $ABG_A G_B$ is convex, and AG_A and BG_B must intersect.



Let AG_A meet BG_B at G . Since $\triangle M_{CD} G_B G_A \sim \triangle M_{CD} AB$, we have $\frac{AB}{G_B G_A} = 3$ and $G_B G_A \parallel AB$. This yields $\triangle G G_A G_B \sim \triangle G AB$, and hence

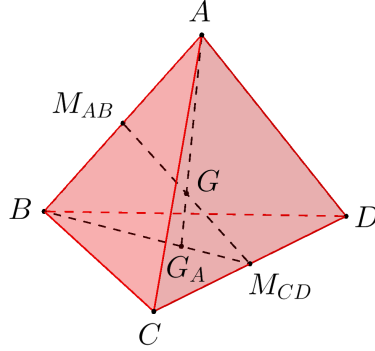
$$\frac{AG}{G G_A} = \frac{BG}{G G_B} = 3.$$

By symmetry, each **median** cuts another **median** in the same ratio $3 : 1$. Therefore, the 4 **medians** are concurrent at G .

(b) Note that AM_{AB} meet $M_{CD} G_A$ at B . Thus, the points A, M_{AB}, G_A, M_{CD} are **coplanar** by proposition 1.4. From the configuration, $AM_{AB} G_A M_{CD}$ is convex, and so AG_A intersects $M_{AB} M_{CD}$. Apply Menelaus' theorem to $\triangle ABG_A$ and the points M_{CD}, G, M_{AB} . As

$$\frac{BM_{CD}}{M_{CD} G_A} \times \frac{G_A G}{G A} \times \frac{AM_{AB}}{M_{AB} B} = (-3) \times \frac{1}{3} \times 1 = -1,$$

the points M_{CD}, G, M_{AB} are collinear. This means the **median** AG_A and the **bimedian** $M_{AB} M_{CD}$ meet at G .



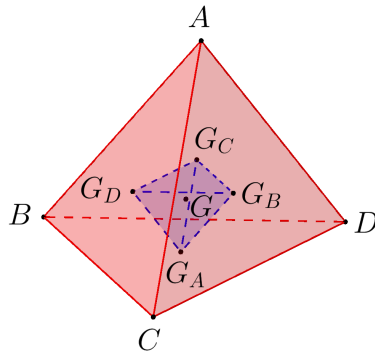
Also, applying Menelaus' theorem to $\triangle M_{AB}BM_{CD}$ and the line AGG_A , we obtain

$$\frac{M_{AB}G}{GM_{CD}} = -\frac{M_{AB}A}{AB} \times \frac{BG_A}{G_AM_{CD}} = -\left(-\frac{1}{2}\right) \times 2 = 1.$$

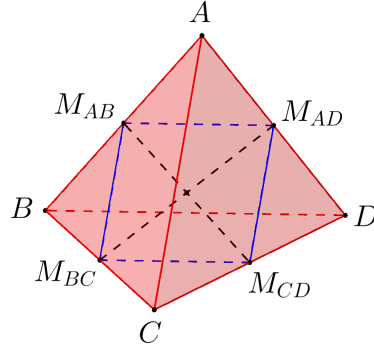
Therefore, G is the midpoint of $M_{AB}M_{CD}$. The proof is complete due to symmetry.

□

A number of corollaries can be obtained from theorem 2.1. Firstly, by the ratios, there exists a homothety mapping $G_AG_BG_CG_D$ to $ABCD$. The centre of the homothety is G , and the ratio is -3 . It follows that the corresponding edges and the corresponding planes are parallel. For example, the plane $G_BG_CG_D$ is parallel to the plane BCD . This can also be seen from the homothety between $AG_BG_CG_D$ and $AM_{CD}M_{BD}M_{BC}$, with centre A and ratio $\frac{3}{2}$. The **centroid** of $G_AG_BG_CG_D$ is the same as the **centroid** of $ABCD$.



Since the segments $M_{AB}M_{CD}$ and $M_{BC}M_{AD}$ bisect each other, $M_{AB}M_{BC}M_{CD}M_{AD}$ forms a parallelogram. This result also holds for the planar case (when $ABCD$ is a degenerate **tetrahedron**), which is in fact a well-known fact in plane geometry called Varignon's theorem(瓦里尼翁定理). In any case, we have $M_{AB}M_{BC} \parallel AC \parallel M_{AD}M_{CD}$ and $M_{AB}M_{AD} \parallel BD \parallel M_{BC}M_{CD}$.



Using the ratio of segments and ratio of volumes, it is easy to prove that

$$[GABC] = [GABD] = [GACD] = [GBCD] = \frac{1}{4}[ABCD].$$

We state one more result without proof. The length of a **median** is given by

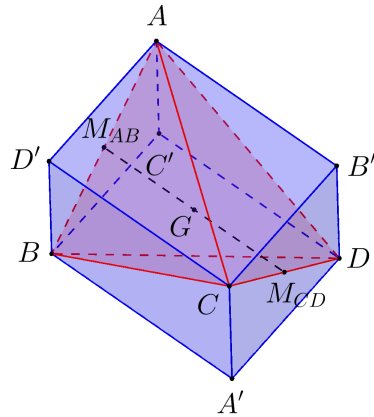
$$AG_A = \sqrt{\frac{AB^2 + AC^2 + AD^2}{3} - \frac{BC^2 + CD^2 + BD^2}{9}}.$$

The only idea is to apply the cosine formula on $\triangle ABG_A$ and $\triangle ABM_{CD}$, and use Apollonius' theorem to compute AM_{CD} and BM_{CD} .

Lastly, we give some results related to the **circumscribed parallelepiped**.

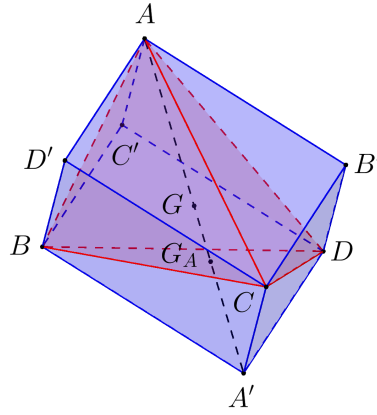
Proposition 2.2. Let $ABCD$ be a **tetrahedron**, and let $AD'CB'DC'BA'$ be its **circumscribed parallelepiped**. Then the following hold.

- (a) The **bimedian** joining the midpoints of the edges AB and CD is parallel to AB' and has the same length as AB' .
- (b) The diagonal AA' contains the **median** from A . Also, the plane BCD cuts AA' in the ratio $2 : 1$, with a longer segment having the endpoint A .



Proof.

- (a) Note that $CA'DB'$ is a parallelogram. So the midpoint M_{CD} of CD is the midpoint of $A'B'$. Also, $ABA'B'$ is a parallelogram. Then the translation in the direction $\overrightarrow{AB'}$ maps the midpoint M_{AB} of AB to the midpoint M_{CD} of $A'B'$, hence the result.
- (b) Recall that the **centroid** G of $ABCD$ is the midpoint of $M_{AB}M_{CD}$. Since we have $AM_{AB} \parallel M_{CD}A'$, the points A, M_{AB}, M_{CD}, A' are **coplanar**. As $AM_{AB} = A'M_{CD}$ and $M_{AB}G = M_{CD}G$, we have $\triangle AM_{AB}G \cong \triangle A'M_{CD}G$. This shows AA' passes through G , so it contains the **median** from A .



Let G_A be the centroid of $\triangle BCD$, which is the intersection of AA' and the plane BCD . From the above congruent triangles, we have $AG = GA'$. Together with $AG : GG_A = 3 : 1$, we easily obtain $AG_A : G_AA' = 2 : 1$ as desired.

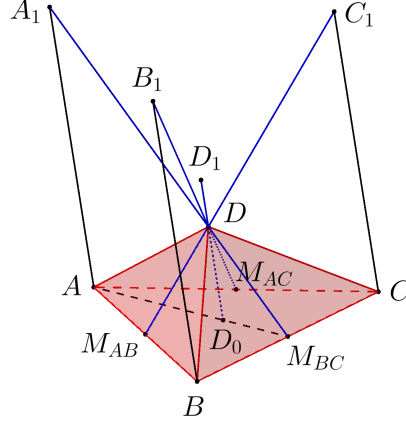
□

Similar results as (a) apply to other **bimedians**. There are also some simple corollaries of (b). The result implies AA', BB', CC', DD' are concurrent at G . Thus, by symmetry, $ABCD$ and $A'B'C'D'$ have the same **centroid** G , and indeed they are homothetic to each other with centre G and ratio -1 . Also, the planes $B'C'D'$ and BCD are parallel and trisect the diagonal AA' . The height from A to BCD is twice the distance from A to the plane $B'C'D'$.

Problem 2.1. (IMO 1964 Problem 6 part 1) In **tetrahedron** $ABCD$, vertex D is connected with D_0 , the centroid of $\triangle ABC$. Lines parallel to DD_0 are drawn through A, B and C . These lines intersect the planes BCD, CAD and ABD in points A_1, B_1 and C_1 , respectively. Prove that the volume of $ABCD$ is one third the volume of $A_1B_1C_1D_0$.

Analysis. The key idea is to locate the point A_1 by considering the intersection of $\overline{A_1D}$ and $\overline{AD_0}$. Afterwards, we have the necessary information about the ratios and can compare the volumes of different **tetrahedrons**.

Solution. As $AA_1 \parallel D_0D$, the points A_1, A, D_0, D are **coplanar**. Since the planes ABC and BCD meet at the line BC , we know that AD_0 intersect A_1D on BC . But we know that AD_0 passes through the midpoint M_{BC} of BC , so the same holds for A_1D . In addition, from $AA_1 \parallel D_0D$, we deduce $A_1D : DM_{BC} = AD_0 : D_0M_{BC} = 2 : 1$. This locates the point A_1 . Similar results hold for B_1 and C_1 .



Now, let D_1 be the point on the ray D_0D beyond D such that $DD_1 = \frac{1}{2}D_0D$.

Then there exists a homothety with centre D and ratio $-\frac{1}{2}$ mapping $A_1B_1C_1D_0$ to $M_{BC}M_{AC}M_{AB}D_1$, where M_{AC} and M_{AB} are the midpoints of AC and AB respectively. It follows that

$$[A_1B_1C_1D_0] = 8[M_{BC}M_{AC}M_{AB}D_1] = 12[M_{BC}M_{AC}M_{AB}D] = 3[ABCD]$$

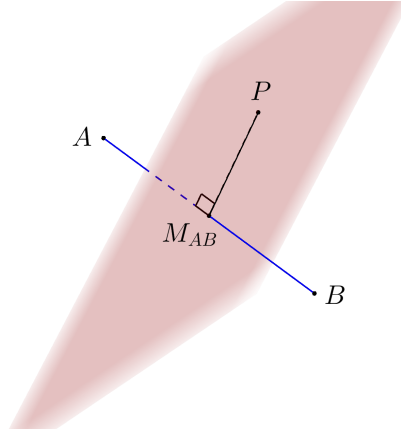
as desired. (The first equality holds due to homothety. The second equality holds by proposition 2.1. The last equality holds as the base area of $\triangle M_{BC}M_{AC}M_{AB}$ is one-fourth of that of $\triangle ABC$.)

2.3 Circumcentre and Circumsphere

In this subsection, we generalize the circumcircle of a triangle in $\mathbb{R}^2$ to the **circumsphere** of a **tetrahedron**. In particular, we will prove that for any **tetrahedron** $ABCD$, there exists a unique **sphere** passing through A, B, C, D . Equivalently, this means there exists a unique **sphere** passing through 4 non-**coplanar** points in $\mathbb{R}^3$. On the other hand, four **coplanar** points are **cospherical** iff they are concyclic, because the intersection of a **sphere** and a plane is a circle.

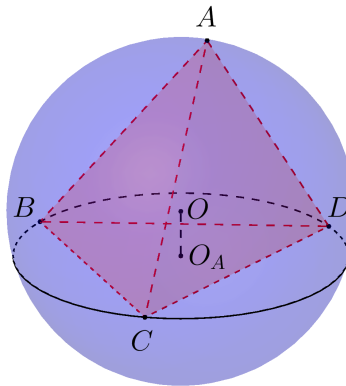
In order to find such a **circumsphere**, its centre O must be equidistant from A, B, C, D . So we need to find the locus of points which are equidistant to two fixed points. The corresponding locus in $\mathbb{R}^2$ is a straight line, the perpendicular bisector of the two fixed points. In $\mathbb{R}^3$, we have a plane.

Definition 2.4. Let A and B be two distinct points in $\mathbb{R}^3$. The *mediating plane*(垂直平分面) or *mediator*(中垂面) of AB is the plane passing through the midpoint of AB and perpendicular to AB .



The **mediator** of AB is the locus of points equidistant to A and B . The proof is simple. Suppose P is a point such that $PA = PB$. If P lies on AB , it must be the midpoint M_{AB} of AB . If P, A, B are non-collinear, then $\triangle PAM_{AB} \cong \triangle PBM_{AB}$. This forces $\angle PM_{AB}A = \angle PM_{AB}B = 90^\circ$, so P lies on the **mediator** of AB . Conversely, for any P lying on this **mediator**, we can show that $\triangle PAM_{AB} \cong \triangle PBM_{AB}$, and hence $PA = PB$.

Using this, we can easily prove the existence and uniqueness of the **circumsphere**. A point O satisfies $OA = OB = OC = OD$ iff it lies on the **mediators** of AB, AC, AD . Since AB, AC, AD do not lie on the same plane, their **mediators** has a unique intersection. Indeed, the **mediators** of AB and AC are not parallel, hence they intersect at a straight line. This straight line is perpendicular to both AB and AC , hence is not perpendicular to AD . Thus, it is not parallel to the **mediator** of AD , and so they have a unique intersection. (This basically says three planes whose **normals** do not lie on the same plane must have a unique intersection.) This gives a unique choice of O and the radius OA .



Definition 2.5. Let $ABCD$ be a **tetrahedron**. The unique **sphere** passing through the vertices A, B, C, D is called the *circumsphere*(外接球) of $ABCD$. The centre O of the **circumsphere** is called the *circumcentre*(外接球球心). The radius of the **circumsphere** is called the *circumradius*(外接球半径).

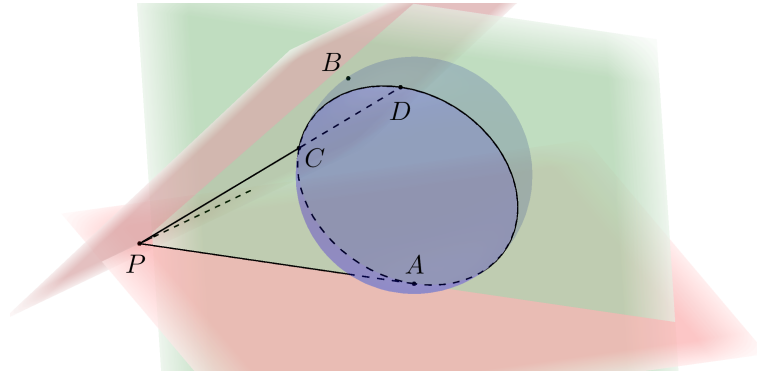
Since O is equidistant to B, C, D , its projection O_A on the plane BCD is also equidistant to B, C, D , and hence O_A is the circumcentre of $\triangle BCD$. In other words, the projection of the centre of a **sphere** to the plane of a **small circle** is the centre of the **small circle**. This is analogous to the planar result: the projection of the centre of a circle to a chord is the midpoint of the chord.

Problems about **circumspheres** are merely problems about **spheres**. They can be solved using similar ideas.

Problem 2.2. (In the World of Mathematics Problem 296) The intersection line of two planes which touch the **circumsphere** of a **tetrahedron** $ABCD$ at points A and B is **coplanar** with the straight line CD . Prove that $\frac{AC}{BC} = \frac{AD}{BD}$.

Analysis. The **coplanar** condition suggests us considering the intersection of the straight line and CD . Then the figure becomes a standard figure of tangent-secant lines.

Solution. Let ℓ be the intersection of the two **tangent planes** to the **circumsphere**, and let O be the **circumcentre**. Since OA and OB are perpendicular to the two planes, the plane OAB is perpendicular to ℓ . If $\ell \parallel CD$, then the plane OAB is perpendicular to CD , and hence OAB is the **mediator** of CD . In this case we have $AC = AD$ and $BC = BD$, so the result holds trivially.



Suppose ℓ and CD are not parallel. Since they are **coplanar**, they intersect at a point P . The plane ACD also contains P , and it cuts the **circumsphere** in a circle Γ . Since PA is **tangent** to the **circumsphere**, it is also tangent to Γ . This shows PA is a tangent to Γ and PCD is a secant to Γ . Thus, $\triangle PAC \sim \triangle PDA$, and hence

$$\frac{AC}{DA} = \frac{PA}{PD} = \frac{PC}{PA} = \sqrt{\frac{PA}{PD} \times \frac{PC}{PA}} = \sqrt{\frac{PC}{PD}}.$$

By symmetry, we also have

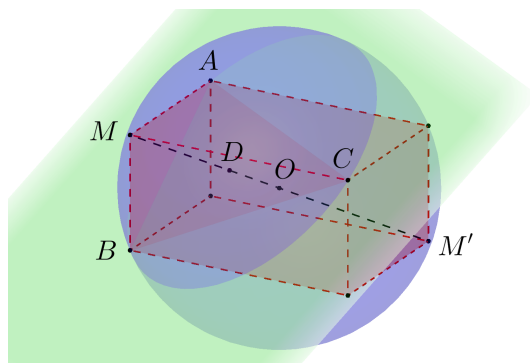
$$\frac{BC}{DB} = \sqrt{\frac{PC}{PD}} = \frac{AC}{DA}.$$

This is the desired relation.

Problem 2.3. (All-Siberian Open School Olympiad 2013 Grade 11 Problem 4) Point M is marked on a **sphere**. Consider all triples of points A, B, C on the **sphere** other than M , such that the segments MA, MB, MC are pairwise perpendicular, and for each such triple we consider the plane passing through A, B, C . Prove that all such planes pass through a common point.

Analysis. If we only focus on the **tetrahedron** $MABC$, it may not be easy to handle the condition $MA \perp MB$, etc. Other than that, we consider a **cuboid** with MA, MB, MC as three sides. This changes the setting to something much easier to work on. Also, due to symmetry, it is not hard to guess that the fixed point should be the intersection of MO and the plane ABC , where O is the centre of the **sphere**.

Solution. Consider the **parallelepiped** with the adjacent sides MA, MB, MC . It is a **cuboid** because MA, MB, MC are pairwise perpendicular. Let M' be the opposite vertex to M of this **cuboid**. Note that the **cuboid** is inscribed in the given **sphere** S , and MM' is a diameter since $\angle MAM' = \angle MBM' = \angle MCM' = 90^\circ$ (this is a simple fact about **cuboids**, analogous to the case of rectangles in $\mathbb{R}^2$). So the midpoint O of MM' is the centre of S .

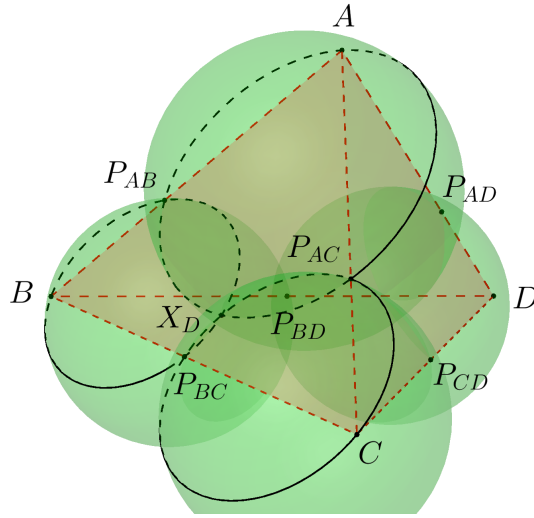


Now, the **cuboid** is the **circumscribed parallelepiped** of $M'ABC$. By proposition 2.2(b), the plane ABC cuts $M'M$ in the ratio $2 : 1$. Therefore, if we let D be the intersection of MM' and plane ABC , then $MD : DO = 2 : 1$. As M and O are fixed, so is D . This means the point D always lies on the plane ABC .

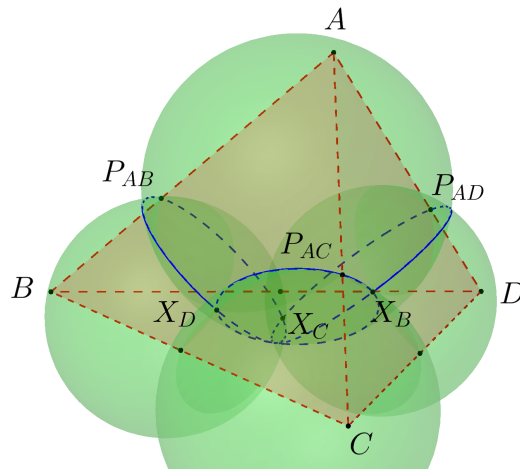
Problem 2.4. Let $ABCD$ be a **tetrahedron**. The points $P_{AB}, P_{AC}, P_{AD}, P_{BC}, P_{BD}$ and P_{CD} are chosen on the edges AB, AC, AD, BC, BD, CD respectively. Prove that the **circumspheres** of $AP_{AB}P_{AC}P_{AD}$, $BP_{AB}P_{BC}P_{BD}$, $CP_{AC}P_{BC}P_{CD}$, $DP_{AD}P_{BD}P_{CD}$ have a common point.

Analysis. This is the analogue of the Miquel point of triangles. The proof is not simple because there is no easy way to prove that 5 points are **cospherical**, in contrast to some simple angle relations for concyclic points. However, just like many other problems in $\mathbb{R}^3$, we may consider some planes and use the corresponding results in $\mathbb{R}^2$ to deduce something.

Solution. We label the **circumsphere** of $AP_{AB}P_{AC}P_{AD}$ as S_A , etc. The plane ABC cuts S_A, S_B, S_C in three circles, which are $(AP_{AB}P_{AC})$, $(BP_{BC}P_{AB})$ and $(CP_{AC}P_{BC})$. These circles meet at the Miquel point X_D of $\triangle P_{BC}P_{AC}P_{AB}$ w.r.t. $\triangle ABC$. Similarly, we can define the Miquel points X_A, X_B, X_C in $\triangle BCD$, $\triangle ACD$, $\triangle ABD$ respectively. Note that X_D lies on S_A, S_B, S_C .



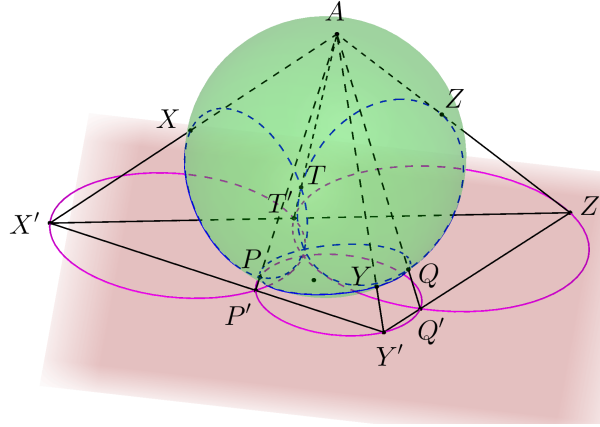
Now, S_A meets S_B, S_C, S_D in $(P_{AB}X_CX_D)$, $(P_{AC}X_BX_D)$, $(P_{AD}X_BX_C)$ respectively. If we can show that these three circles have a common point, then this common point lies on all of S_A, S_B, S_C, S_D as desired. The question is reduced to the following (with $X = P_{AB}$, $Y = P_{AC}$, $Z = P_{AD}$, $P = X_D$, $Q = X_B$, $T = X_C$), which we state as a claim.



Claim. Suppose $AXYZ$ is a fixed **tetrahedron**. We choose any points P, Q, T on (AXY) , (AYZ) , (AZX) respectively. Then (XTP) , (YPQ) , (ZQT) must have a common point.

To prove this claim, we use another technique called the *stereographic projection* (球極平面投影). This means to project every other point on a **sphere** S from a point A on S to the plane Π **tangent** to the **antipode** of A , i.e. the image of $P \in S$ is the unique point $P' \in \Pi$ such that A, P, P' are collinear. It is easy to show that this mapping is bijective (between $S - \{A\}$ and Π) and continuous.

This mapping behaves very similar to inversion (in fact, it is a special case of an inversion w.r.t. a **sphere**). We state a few properties without proof. Circles on S that do not pass through A are projected to circles on Π . Circles on S that pass through A are projected to lines on Π . The mapping preserves angles.

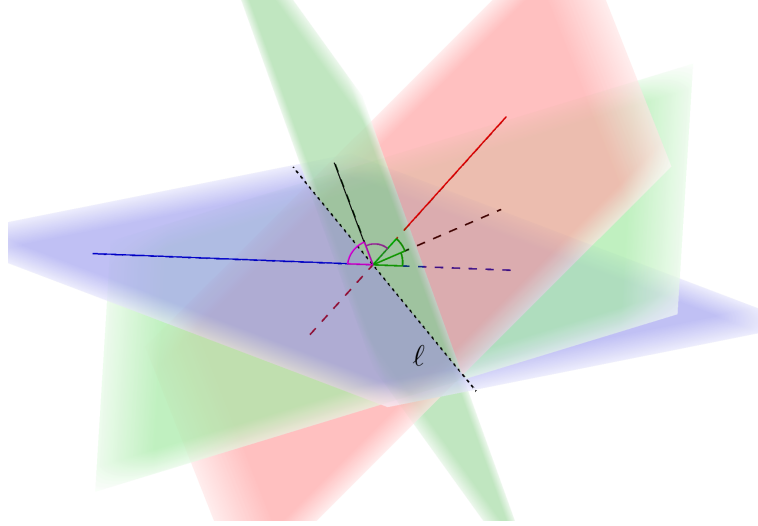


We return to the proof of the claim. Note that P, Q, T lie on the **circumsphere** of $AXYZ$. Let X', Y', Z', P', Q', T' be the images of X, Y, Z, P, Q, T under the **stereographic projection** from A . The circle $(AXPY)$ is mapped to a straight line, meaning that P' lies on $X'Y'$. Similarly, Q' lies on $Y'Z'$, and T' lies on $Z'X'$. Thus, $(X'T'P')$, $(Y'P'Q')$, $(Z'Q'T')$ meet at the Miquel point M of $\triangle P'Q'T'$ w.r.t. $\triangle X'Y'Z'$. The preimages of these three circles are (XTP) , (YPQ) , (ZQT) respectively. Hence, these three circles meet at a point, which is the preimage of M . This proves the claim and solves the whole problem.

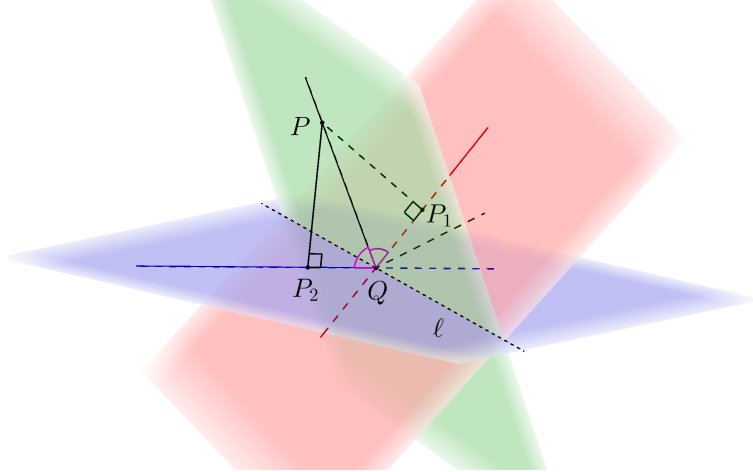
2.4 Insphere and Exspheres

The **insphere** and the **exspheres** are the **spheres** which are **tangent** to the planes of the 4 faces of a **tetrahedron**. To determine these **spheres**, note that the distances from their centres to the 4 faces are the same. So we first consider the locus of points which have the same distance to two non-parallel planes.

Definition 2.6. Let Π_1 and Π_2 be two non-parallel planes in $\mathbb{R}^3$ intersecting in the line ℓ . The *bisecting planes*(二面角平分面) of the **dihedral angle** between Π_1 and Π_2 are the planes that contain ℓ and bisect the **dihedral angle**.



For each pair of non-parallel planes, there are two **bisecting planes**, which are perpendicular to each other. For a **tetrahedron** $ABCD$, the **bisecting planes** w.r.t. AB mean the **bisecting planes** of the **dihedral angle** between the planes ABC and ABD . One of these **bisecting planes** cuts the interior of $ABCD$, and hence is called the *internal bisecting plane*. The other **bisecting plane** lies outside $ABCD$, and hence is called the *external bisecting plane*.

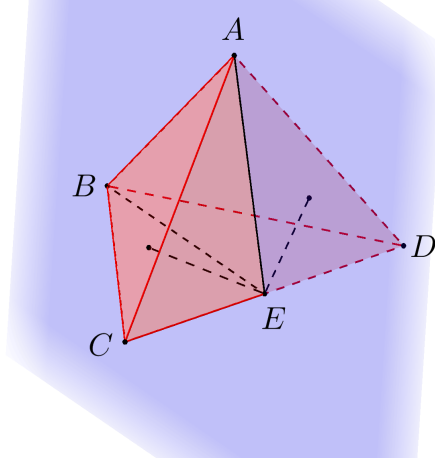


The two **bisecting planes** of the **dihedral angle** between Π_1 and Π_2 form the locus of points which are equidistant to Π_1 and Π_2 . Indeed, if P is such a point, let P_1 and P_2 be the projections of P on Π_1 and Π_2 respectively. Let Q be the intersection of the plane PP_1P_2 and ℓ . Then $\triangle PQP_1 \cong \triangle PQP_2$, and so PQ bisects $\angle P_1QP_2$. This shows P lies on a **bisecting plane**. The converse can be proved by noting the same pair of congruent triangles.

Analogous to the angle bisector theorem, we have the following result.

Proposition 2.3. Let E be the intersection of the line CD and a **bisecting plane** w.r.t. AB . Then

$$\frac{CE}{DE} = \frac{[ABC]}{[ABD]}.$$



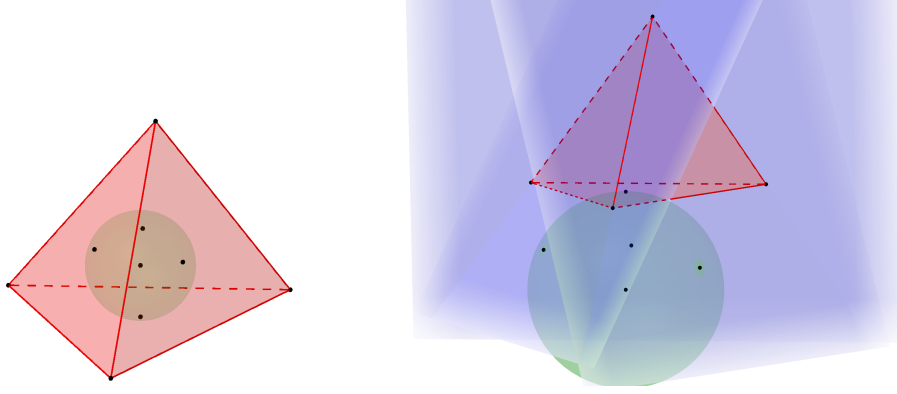
Proof. Since E lies on a **bisecting plane** w.r.t. AB , it is equidistant to the planes ABC and ABD . Therefore, the ratio of volumes of $ABCE$ and $ABDE$ is equal to the ratio of areas of bases ABC and ABD . Thus,

$$\frac{[ABC]}{[ABD]} = \frac{[ABCE]}{[ABDE]} = \frac{CE}{DE}.$$

The last equality follows from proposition 2.1. Note that the proof works for both **internal** and **external bisecting planes**. $\square$

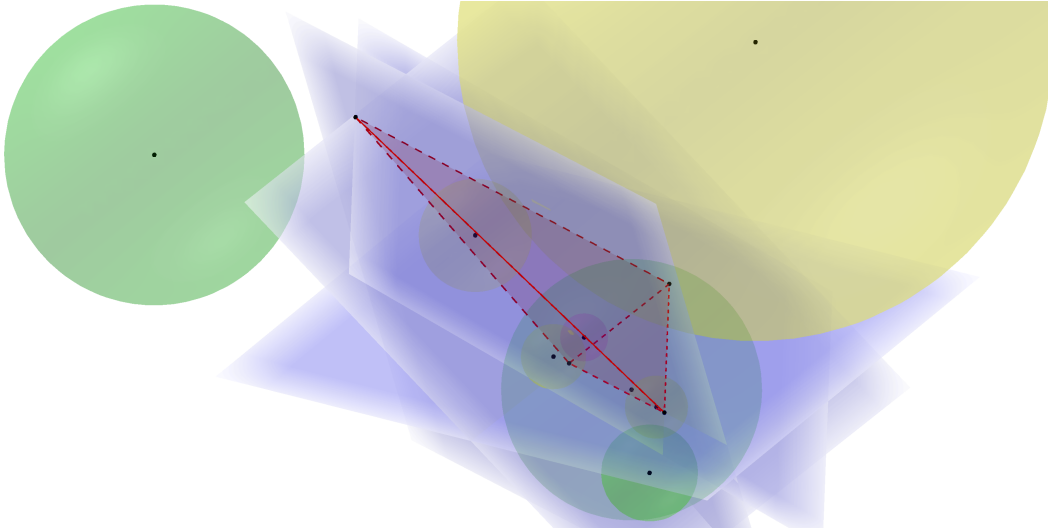
Continuing the above discussion, we know that the centre of a **sphere tangent** to the 4 faces of a **tetrahedron** $ABCD$ must lie on all the **bisecting planes** w.r.t. 6 edges. Indeed, it suffices for the point to lie on the **bisecting planes** w.r.t. AB , AC and BD . For each choice (internal or external) of the **bisecting plane** w.r.t. each edge, there exists at most one intersection point. Therefore, there are at most $2^3 = 8$ points which are equidistant to the 4 faces. At each such point, a **sphere** touching the 4 faces can be drawn. This gives the following definitions.

Definition 2.7. Let $ABCD$ be a **tetrahedron**. There exists a unique **sphere** lying inside $ABCD$ which is **tangent** to the 4 faces of $ABCD$. It is called the *insphere*(内切球) of $ABCD$, and its centre is called the *incentre*(内切球球心). The radius of the **insphere** is called the *inradius*(内切球半径).



Definition 2.8. Let $ABCD$ be a **tetrahedron**. There exists a unique **sphere** externally **tangent** to the face BCD and internally **tangent** to the other 3 faces of $ABCD$. It is called the *exsphere*(旁切球) of $ABCD$ opposite to A , and its centre is called an *excentre*(旁切球球心).

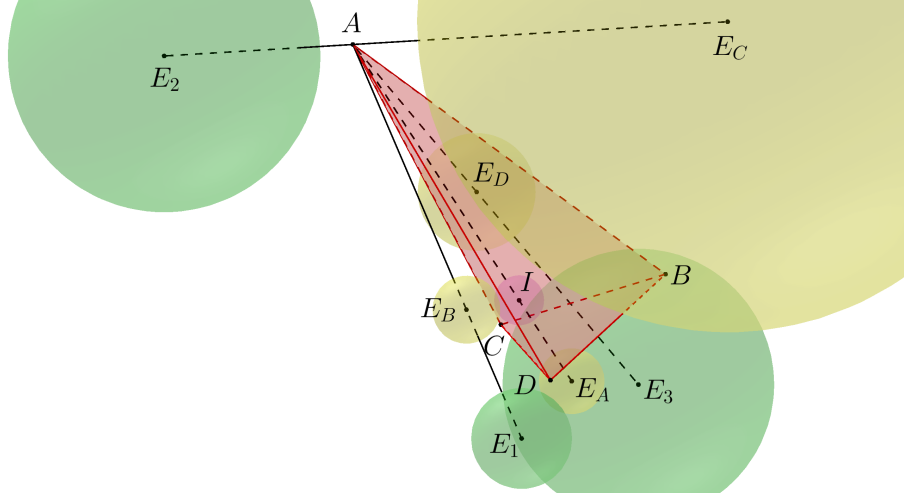
On the other hand, there exists at most one **sphere** lying on the **internal bisecting planes** w.r.t. AB and CD , and the **external bisecting planes** w.r.t. the other 4 edges. It is called the *exsphere* of $ABCD$ opposite to AB and CD , and its centre is called an *excentre*.



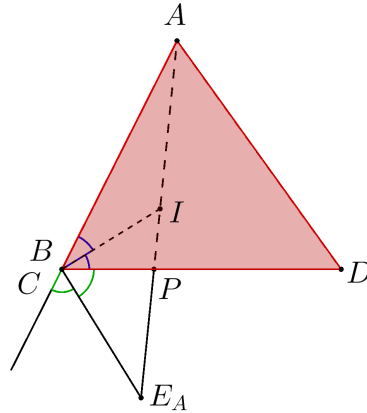
There are many collinear points in the figure. Let I be the **incentre**, E_A, E_B, E_C, E_D the **excentres** opposite to A, B, C, D , and E_1, E_2, E_3 the **excentres** opposite to the pairs AB, CD ; AC, BD ; AD, BC respectively. By considering the intersection of different **bisecting planes**, we can show that the following groups of points are collinear.

$$\begin{array}{cccc}
 (A, I, E_A) & (A, E_B, E_1) & (A, E_C, E_2) & (A, E_D, E_3) \\
 (B, I, E_B) & (B, E_A, E_1) & (B, E_D, E_2) & (B, E_C, E_3) \\
 (C, I, E_C) & (C, E_D, E_1) & (C, E_A, E_2) & (C, E_B, E_3) \\
 (D, I, E_D) & (D, E_C, E_1) & (D, E_B, E_2) & (D, E_A, E_3)
 \end{array}$$

In addition, the vertex A, B, C, D in each group is the centre of homothety mapping one **insphere/exsphere** to the other.

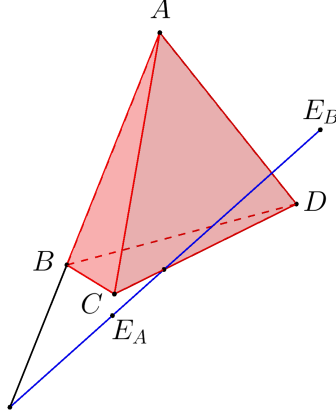


Let AIE_A meet the plane BCD at P . As in the planar case, $(A, P; I, E_A)$ is a harmonic division. In fact, we choose a plane with BC as a **normal**, and project all the points on this plane. We use the ambiguous notation that the image of each point has the same label as that point.

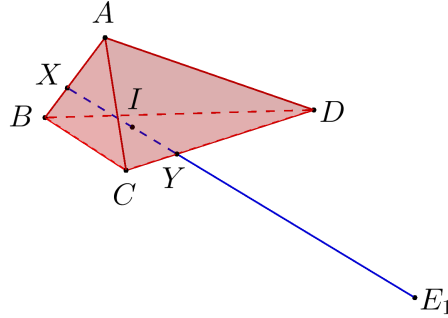


After the projection, the points B and C overlap. Since I and E_A lie on the **internal** and **external bisecting planes** w.r.t. BC , their projections lie on the internal and external angle bisectors of $\angle DBA$. Also, P is projected to the intersection of AIE_A and BD . It follows from a well-known fact that $(A, P; I, E_A) = -1$. As projection preserves cross ratio, the same holds for the four points before projection.

The **excentres** E_A and E_B lie on the **internal bisecting plane** w.r.t. AB , and the **external bisecting plane** w.r.t. CD , so both of them lie on the intersection line of these two planes. Thus, this line is **coplanar** with the line AB and also with the line CD . In other words, $E_A E_B$ either intersects AB or is parallel to AB , and the same holds with AB replaced by CD .



Similar results hold for other **excentres**, including E_1, E_2, E_3 . For example, both I and E_1 lie on the **internal bisecting planes** w.r.t. AB and CD , and so IE_1 is the intersection of these planes. Note that these planes must cut the interior of the opposite edges, and so their intersection line must meet the interior of AB and CD at some points X and Y respectively (i.e. the parallel case never holds). We can also show that $(X, Y; I, E_1) = -1$, again by projecting the figure on a plane with **normal** BC .



We can deduce some more facts from the harmonic division. Consider the **tetrahedron** $XBCY$. The **internal bisecting plane** w.r.t. BC in this **tetrahedron** coincides with that in $ABCD$. Thus, we can apply proposition 2.3 to obtain

$$\frac{IX}{IY} = \frac{[BCX]}{[BCY]}.$$

Similarly, by considering the **tetrahedrons** $XDBY$, $XCAY$ and $XDAY$, we obtain

$$\frac{IX}{IY} = \frac{[BCX]}{[BCY]} = \frac{[BDX]}{[BDY]} = \frac{[ACX]}{[ACY]} = \frac{[ADX]}{[ADY]} = k$$

for some constant k . It follows that

$$\begin{aligned} [ABC] + [ABD] &= [BCX] + [ACX] + [BDX] + [ADX] \\ &= k[BCY] + k[ACY] + k[BDY] + k[ADY] \\ &= k([BCD] + [ACD]). \end{aligned}$$

Thus, $\frac{IX}{IY} = \frac{[ABC] + [ABD]}{[BCD] + [ACD]}$. Since $(X, Y; I, E_1)$ forms a harmonic division, we have $\frac{E_1X}{E_1Y} = \frac{[ABC] + [ABD]}{[BCD] + [ACD]}$.

However, it is possible that $[ABC] + [ABD] = [BCD] + [ACD]$. In that case, we need $E_1X = E_1Y$, which is impossible as E_1 lies outside the segment XY (as I lies on the segment XY). In other words, the **excentre** E_1 (hence the corresponding **exsphere**) does not exist when the above area relation holds. Similar results hold for the existence of E_2 and E_3 .

If both $[ABC] + [ABD] = [BCD] + [ACD]$ and $[ABC] + [ACD] = [ABD] + [BCD]$ hold, i.e. $[ABC] = [BCD]$ and $[ABD] = [ACD]$, both E_1 and E_2 do not exist. It can be shown that this is equivalent to $AB = CD$ and $AC = BD$.

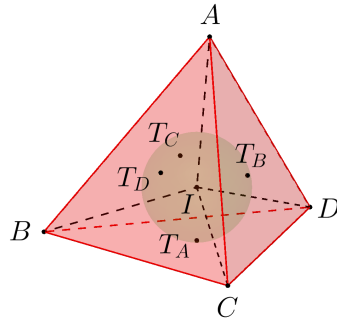
If $[ABC] = [ABD] = [ACD] = [BCD]$, i.e. $AB = CD$, $AC = BD$ and $AD = BC$, all of E_1, E_2, E_3 do not exist. It follows that a **tetrahedron** may have exactly 4, 5, 6 or 7 **excentres**.

Next, we give some more properties related to the **incentre** and the **insphere**. In many cases, there are some analogous results for the **excentres**.

Let the **insphere** touch the faces ABC, ABD, ACD, BCD at T_D, T_C, T_B, T_A respectively. The volume $[IABC]$ is equal to $\frac{1}{3}r[ABC]$, where r is the **inradius** of $ABCD$. This is because the radius IT_D is perpendicular to the plane ABC . Similar formulae hold for the other **tetrahedrons**. Thus,

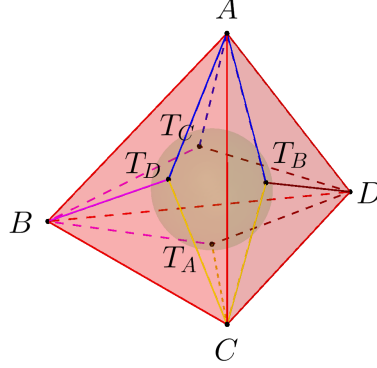
$$\begin{aligned} [ABCD] &= [IABC] + [IABD] + [IACD] + [IBCD] \\ &= \frac{1}{3}r([ABC] + [ABD] + [ACD] + [BCD]). \end{aligned}$$

This is similar to the formula $[ABC] = sr$ for the planar case.

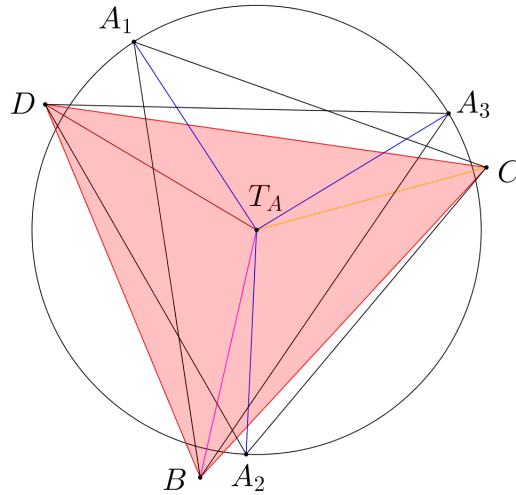


Since the **tangents** from the same point to the **insphere** have the same length, we have $AT_B = AT_C = AT_D$, and similarly for other **tangents** from different vertices. These generate 6 pairs of congruent triangles in the figure, such as $\triangle ABT_C \cong \triangle ABT_D$.

Many properties of the tangential points can be deduced using these congruent triangles.



As an application, we provide the following construction of the tangential points on each plane. For example, let us focus on the plane BCD .

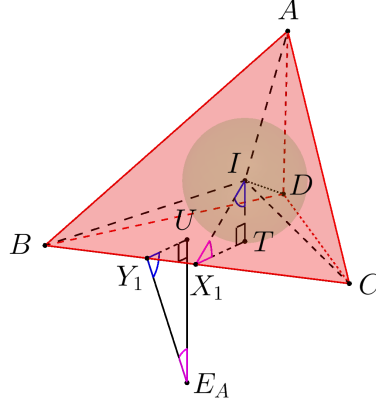


Let A_1 be the point (on the plane BCD) such that $\triangle A_1BC \cong \triangle ABC$, and A_1, D lie on the same side of BC . Let A_2 be the point such that $\triangle A_2CD \cong \triangle ACD$, and A_2, B lie on the same side of CD . Let A_3 be the point such that $\triangle A_3DB \cong \triangle ADB$, and A_3, C lie on the same side of DB .

Note that $ABCT_D$ is congruent to A_1BCT_A , which implies $AT_D = A_1T_A$. Similarly, $AT_B = A_2T_A$ and $AT_C = A_3T_A$. As all these **tangents** have the same length, we have $A_1T_A = A_2T_A = A_3T_A$. Therefore, T_A is the circumcentre of $\triangle A_1A_2A_3$. This gives the construction of T_A using only information of the vertices A, B, C, D .

Similar constructions work for the tangential points of the **exspheres** with the plane of a face. The main difference is that we need to choose each triangle A_1BC , etc., carefully because there are two such triangles congruent to $\triangle ABC$, and only one has the same orientation that maps a tangential point to a tangential point.

We prove one more result relating the contact points of the **insphere** and the **A-exsphere** with the plane BCD . For simplicity, we write T is the contact point of the **insphere** and the plane BCD . Let U be the contact point of the **A-exsphere** and the plane BCD . Then T and U are isogonal conjugate of each other w.r.t. $\triangle BCD$.



Let X_1, X_2, X_3 be the projection of T on the sides BC, CD, DB , and let Y_1, Y_2, Y_3 be the projection of U on the sides BC, CD, DB respectively. As I and E_A lie on different **bisecting planes** w.r.t. BC , we have

$$\angle TX_1I + \angle E_A Y_1U = \frac{1}{2} \cdot 180^\circ = 90^\circ.$$

Thus, the right-angled triangles $\triangle ITX_1$ and $\triangle Y_1UE_A$ are similar. From this, we obtain $\frac{IT}{TX_1} = \frac{Y_1U}{UE_A}$, i.e. $TX_1 \times Y_1U = IT \times UE_A = rr_A$, where r and r_A are the radii of the **insphere** and the **A-exsphere** respectively. By symmetry, we have

$$TX_1 \times UY_1 = TX_2 \times UY_2 = TX_3 \times UY_3 = rr_A.$$

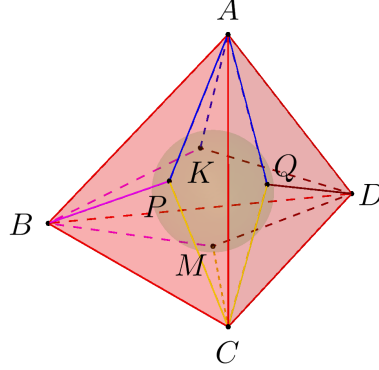
This implies T and U are isogonal conjugate w.r.t. $\triangle BCD$ by a well-known result.

Problem 2.5. (IMO Longlist 1985) The **sphere** inscribed in **tetrahedron** $ABCD$ touches the sides ABD and DBC at points K and M , respectively. Prove that $\angle AKB = \angle DMC$.

Analysis. Recall that there are 6 pairs of congruent triangles related to the tangential points. We can use these to find the relationship between the two angles.

It may not be easy to compare the angles in a three-dimensional figure, even if the figure is perfect (for example, drawn by software). One suggestion is to draw those triangles as some separate figures on the plane, or draw the triangles on the same face in the same figure, and mark the equal angles. After doing so (and knowing the congruent triangles), this problem becomes a trivial one. The only facts we need are angle sum of triangle and angle at a point.

Solution.



Let P and Q be the contact points of the **insphere** and the planes ABC and ACD respectively. Recall that $\triangle ABK \cong \triangle ABP$, $\triangle ACP \cong \triangle ACQ$, $\triangle ADQ \cong \triangle ADK$, $\triangle BCP \cong \triangle BCM$, $\triangle BDK \cong \triangle BDM$, $\triangle CDQ \cong \triangle CDM$. Therefore, we have

$$\begin{aligned}
 2\angle AKB &= \angle AKB + \angle APB \\
 &= (360^\circ - \angle AKD - \angle BKD) + (360^\circ - \angle APC - \angle BPC) \\
 &= (360^\circ - \angle AQD - \angle BMD) + (360^\circ - \angle AQC - \angle BMC) \\
 &= (360^\circ - \angle AQD - \angle AQC) + (360^\circ - \angle BMD - \angle BMC) \\
 &= \angle CQD + \angle CMD = 2\angle CMD.
 \end{aligned}$$

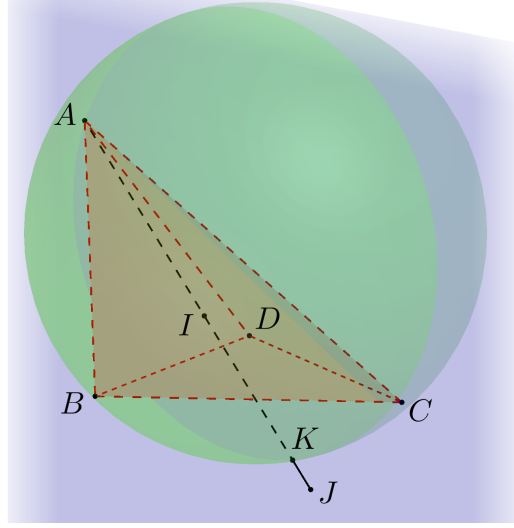
This yields $\angle AKB = \angle CMD$.

Problem 2.6. (Sharygin Geometry Olympiad 2020 Correspondence Problem 24) Let I be the **incentre** of a **tetrahedron** $ABCD$, and J be the centre of the **exsphere** touching the face BCD and the planes containing three remaining faces (outside these faces). The segment IJ meets the **circumsphere** of the **tetrahedron** at point K . Which of two segments IK and JK is longer?

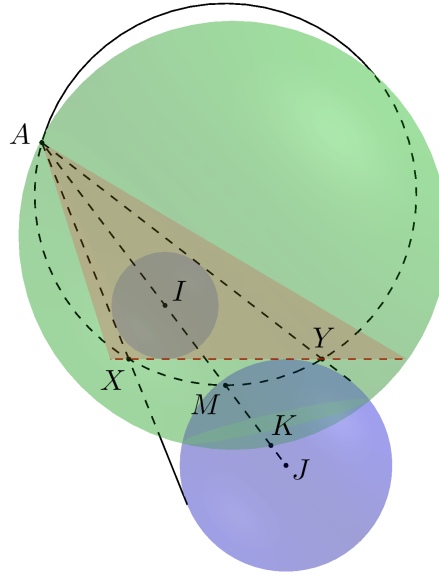
Analysis. In the $\mathbb{R}^2$ case, we always have $IK = JK$. However, this does not seem to be the case in $\mathbb{R}^3$ by considering some special cases. Yet we can still reduce the problem to the $\mathbb{R}^2$ case by considering the intersection of a plane with the figure, and see how this is related to the original problem.

The choice of the plane is a crucial step in many such problems. For problems involving **spheres**, it is always useful to consider a plane passing through the centres. Therefore, our first priority is to consider a plane containing the points I and J . There are infinitely many such planes, but it seems that there is not much difference. So we follow the official solution and make a natural choice below.

Solution. Consider the plane passing through AIJ and perpendicular to the plane BCD . This plane passes through the centres I and J of the **insphere** and the **exsphere**, hence it cuts these **spheres** in their **great circles**.



Also, as A is the external centre of similitude of the **insphere** and the **exsphere**, it is also the external centre of similitude of the **great circles**. Let the two common tangents of the **great circles** from A meet the plane BCD at X and Y respectively. Then the **great circles** are the incircle and excircle of $\triangle AXY$. So the midpoint M of IJ is the midpoint of $\widehat{XY}$ on (AXY) opposite to A .



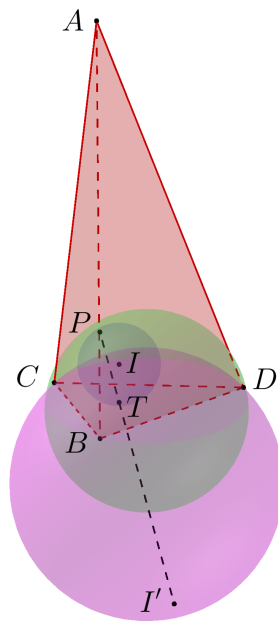
Now, the above plane cannot pass through two of the edges AB, AC, AD . So the incircle of $\triangle AXY$ must lie inside the triangle formed by the intersection of the plane and $ABCD$, possibly touching one side of the triangle from A but not both. This shows one of X and Y lies inside $\triangle BCD$, and the other lies on the boundary of or inside $\triangle BCD$. In any case, $\widehat{XY}$ not containing A must lie strictly inside the **circumsphere** of $ABCD$. Therefore, M lies on the segment IK , and

$$IK > IM = MJ > KJ.$$

Problem 2.7. Let I be the incentre of a tetrahedron $ABCD$. The insphere touches the plane BCD at T . Let I' be the antipode of I on the circumsphere of $IBCD$. The line $I'T$ meets the insphere of $ABCD$ again at P . Prove that the circumsphere of $PBCD$ is tangent to the insphere of $ABCD$.

Analysis. Although the problem looks even more tedious, we can use the same trick as the previous problem, i.e. to consider some plane containing the centres of the spheres, and then work on the corresponding problem in $\mathbb{R}^2$. The $\mathbb{R}^2$ version of this problem is also a good exercise to work on.

Solution.



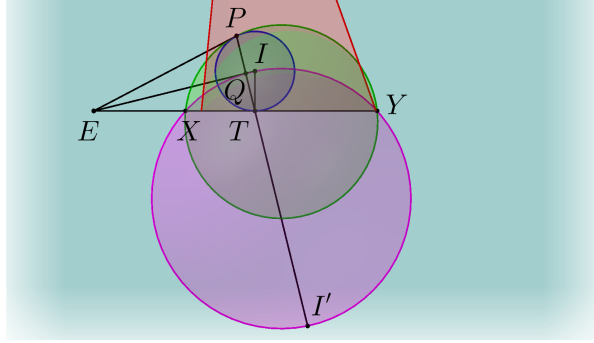
Consider the plane Π passing through II' and perpendicular to the plane BCD . Note that T also lies on Π because T is the projection of I on the plane BCD . It follows that P lies on Π as well.

Π passes through the centre of the insphere and the circumcentre of $IBCD$ (which is the midpoint of II'), hence it cuts these spheres in their great circles Γ_1 and Γ_2 . In particular, II' is still a diameter of Γ_2 .

On the other hand, let Π cut the circumsphere of $PBCD$ at a circle Γ_3 . The intersection points X, Y of Γ_2 and Γ_3 must lie on the intersection of the circumspheres of $IBCD$ and $PBCD$, which is (BCD) . Therefore, XY is the intersection of Π and the plane BCD , and hence T lies on XY .

It is not hard to see that the circumsphere of $PBCD$ is tangent to the insphere of $ABCD$ iff Γ_3 is tangent to Γ_1 . So we can now work on this planar problem.

Let $I'T$ meet Γ_2 again at Q . Then $\angle I'QI = 90^\circ$ since II' is a diameter of Γ_2 . If T lies on II' , then the figure is symmetric about the line II' , and the result clearly holds.



Otherwise, IQ and XY meet at some point E . Note that IQE is the perpendicular bisector of TP .

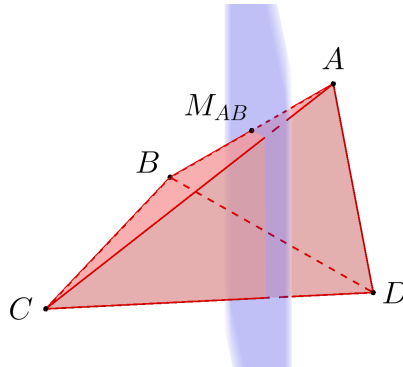
As $\angle EQT = \angle ETI = 90^\circ$, we have $\triangle EQT \sim \triangle ETI$, and hence $ET^2 = EQ \times EI$. This yields

$$EP^2 = ET^2 = EQ \times EI = EX \times EY.$$

Therefore, EP is tangent to Γ_3 . Also, from $ET^2 = EX \times EY$, the point E has the same power w.r.t. $\Gamma_1, \Gamma_2, \Gamma_3$. This implies EP is the radical axis of Γ_1 and Γ_3 . Thus, Γ_1 and Γ_3 are tangent to each other. This completes the proof.

2.5 Monge Point and Twelve-point Sphere

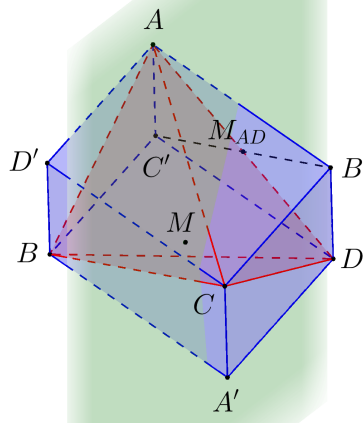
So far we have not mentioned any centre analogous to the orthocentre of a triangle. The main reason is that the **altitudes** of the **tetrahedron** need not meet at a point. A special kind of **tetrahedron** whose **altitudes** are concurrent will be discussed in section 3. For general **tetrahedrons**, we have a substituted centre called the **Monge point**, which shares some common properties analogous to the orthocentre.



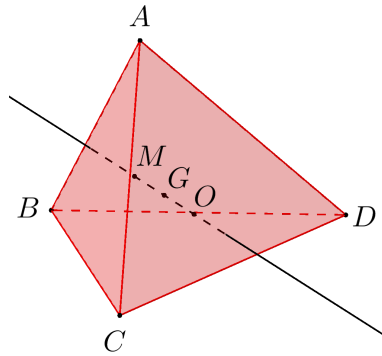
Definition 2.9. Let $ABCD$ be a **tetrahedron**. A *Monge plane*(蒙日平面) is a plane passing through the midpoint of one edge of $ABCD$ and perpendicular to the opposite edge.

For example, the plane passing through the midpoint of AB and perpendicular to CD is a **Monge plane**. We now prove the following important result about the **Monge planes** of $ABCD$.

Theorem 2.2. (Monge's theorem(蒙日定理)) Let $ABCD$ be a **tetrahedron**. The 6 **Monge planes** of $ABCD$ meet at a point M , called the *Monge point*(蒙日點) of $ABCD$.

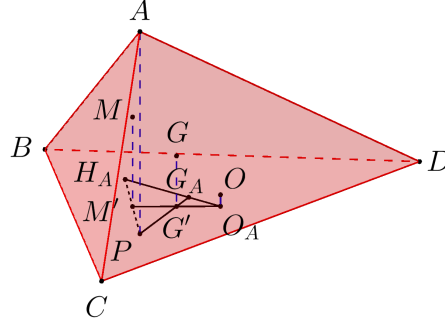


Proof. Let $AD'CB'DC'BA'$ be the **circumscribed parallelepiped** of $ABCD$. Consider the **Monge plane** passing through the midpoint M_{AD} of AD . Since $AC'DB'$ is a parallelogram, M_{AD} is also the midpoint of $C'B'$. Also, BC is parallel to $C'B'$. So the **Monge plane** is simply the **mediator** of $B'C'$. Similarly, the 6 **Monge planes** of $ABCD$ are the **mediators** of the 6 edges of $A'B'C'D'$. They intersect at the **circumcentre** of $A'B'C'D'$. $\square$



From the above proof, we easily deduce some more facts related to the **Monge point**. For example, the **Monge point** of $ABCD$ is the **circumcentre** of $A'B'C'D'$. Since $ABCD$ and $A'B'C'D'$ are congruent and are homothetic with centre G and ratio -1 , the **circumcentre** of $ABCD$ is mapped to the **circumcentre** of $A'B'C'D'$ under the homothety. So G is the midpoint of OM . As in the planar case, the line OGM is called the *Euler line*(歐拉線) of $ABCD$.

Let O_A, G_A, H_A be the circumcentre, centroid and orthocentre of $\triangle BCD$, and let P, M', G' be the projection of A, M, G on the plane BCD respectively. Recall that O_A is the projection of O on the plane BCD . We shall prove that M' is the midpoint of H_A and P .



We have the following ratio of the collinear points.

- H_A, G_A, O_A are collinear with $H_A G_A : G_A O_A = 2 : 1$ by the Euler line of $\triangle BCD$
- M', G', O_A are collinear with $M' G' : G' O_A = 1 : 1$ since G is the midpoint of MO
- P, G', G_A are collinear with $P G' : G' G_A = 3 : 1$ since A, G, G_A are collinear with $AG : G G_A = 3 : 1$

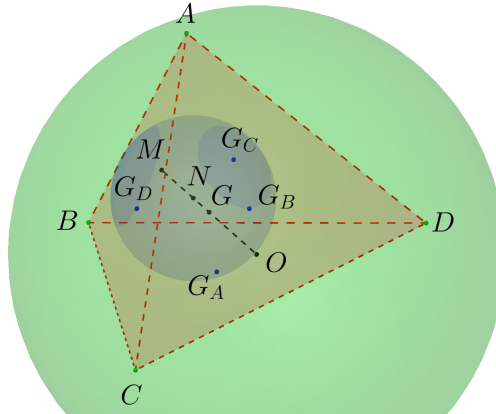
Using these, we obtain

$$\frac{G_A H_A}{H_A O_A} \times \frac{O_A M'}{M' G'} \times \frac{G' P}{P G_A} = \frac{-2}{3} \times \frac{-2}{1} \times \frac{-3}{4} = -1.$$

By Menelaus' theorem, H_A, M', P are collinear. Also, we have

$$\frac{M' P}{P H_A} = -\frac{M' G'}{G' O_A} \times \frac{O_A G_A}{G_A H_A} = -\frac{1}{1} \times \frac{1}{2} = -\frac{1}{2},$$

implying M' is the midpoint of $H_A P$. As a corollary, the **Monge point** lies on the plane containing H_A and the **A-altitude**.



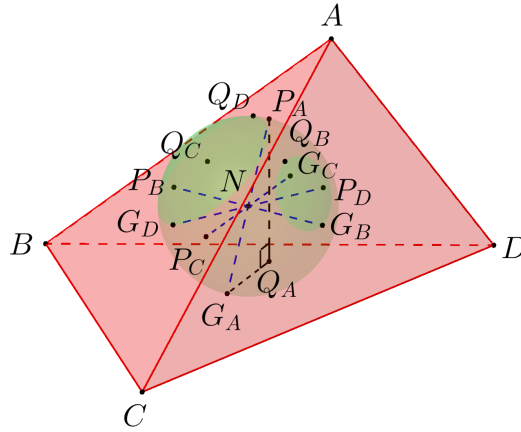
Next, we give an analogue of the nine-point circle of triangles. Recall that the nine-point circle is the circumcircle of the 3 midpoints of the sides of a triangle, which can also be obtained by a homothety of the circumcircle with the orthocentre as the centre and ratio $\frac{1}{2}$. We have a similar result using the **Monge point** in replace of the orthocentre, and with a change of the ratio.

Let G_A, G_B, G_C, G_D be the centroids of $\triangle BCD, \triangle ACD, \triangle ABD, \triangle ABC$ respectively. The **centroid** G of $ABCD$ is the homothetic centre between $ABCD$ and $G_A G_B G_C G_D$. The ratio of this homothety is $-\frac{1}{3}$. This homothety maps the **circumsphere** of $ABCD$ to that of $G_A G_B G_C G_D$. In particular, the centre N of the **circumsphere** of $G_A G_B G_C G_D$ lies on the line GO such that $NG : GO = 1 : 3$ (with G between N and O).

On the other hand, since $MG = GO$, we have $MN : NO = 1 : 2$, i.e. $NM : MO = -1 : 3$. So M is the external centre of similitude between the two **spheres**. This gives the following definition.

Definition 2.10. Let $ABCD$ be a **tetrahedron**. The *twelve-point sphere*(十二點球) of $ABCD$ is the image of the **circumsphere** under the homothety with centre M , the **Monge point** of $ABCD$, and ratio $\frac{1}{3}$. The centre of this **sphere** is called the *twelve-point centre*(十二點球心) of $ABCD$.

From its name, there should be 12 special points lying on the **twelve-point sphere**. We have already mentioned the points G_A, G_B, G_C, G_D . Here are some other points.



Let P_A, P_B, P_C, P_D be the points on the segments MA, MB, MC, MD such that

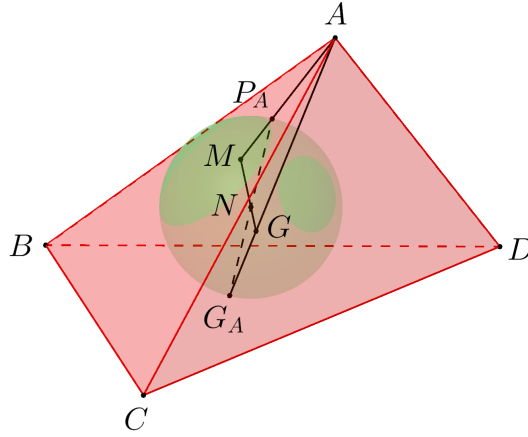
$$AP_A : P_A M = BP_B : P_B M = CP_C : P_C M = DP_D : P_D M = 2 : 1.$$

By the homothety, the points P_A, P_B, P_C, P_D lie on the **twelve-point sphere**.

Consider $\triangle AMG$. The points P_A, N, G_A lie on the sidelines, and we have

$$\frac{AP_A}{P_A M} \times \frac{MN}{NG} \times \frac{GG_A}{G_A A} = \frac{2}{1} \times \frac{2}{1} \times \frac{-1}{4} = -1.$$

By Menelaus' theorem, the points P_A, N, G_A are collinear. This shows $P_A G_A$ is a diameter of the **twelve-point sphere**. Therefore, if we define Q_A to be the projection of P_A on the plane BCD , so that $\angle G_A Q_A P_A = 90^\circ$, the point Q_A also lies on the **twelve-point sphere**. Similarly, the projection Q_B, Q_C, Q_D of P_B, P_C, P_D on the opposite faces lie on the **twelve-point sphere**. This gives a total of 12 points, G_A, P_A, Q_A and their symmetric points.



Unlike the case in $\mathbb{R}^2$, the **twelve-point sphere** and the **insphere** do not touch each other in general.

Problem 2.8. Given a **tetrahedron** $ABCD$, let the perpendicular line from A to the plane BCD meet the **circumsphere** of $ABCD$ again at A_1 . Let A_2 be the reflection of A_1 in the plane BCD . Prove that the centroids of $\triangle A_2 BC$, $\triangle A_2 BD$ and $\triangle A_2 CD$ lie on the **twelve-point sphere** of $ABCD$.

Analysis. This is the same as proving the **twelve-point sphere** of $A_2 BCD$ coincides with that of $ABCD$, because both of them contain the centroid of $\triangle BCD$. We prove separately that they have the same centre and radius.

Solution. Let O, G and N be the **circumcentre**, **centroid** and **twelve-point centre** of $ABCD$. Let G_1 and N_1 be the **centroid** and **twelve-point centre** of $A_1 BCD$, and let G_A be the centroid of $\triangle BCD$.

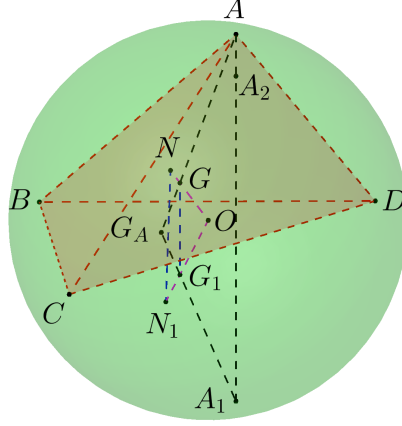
The points G and G_1 lie on AG_A and $A_1 G_A$ such that

$$AG : GG_A = A_1 G_1 : G_1 G_A = 3 : 1.$$

Since AA_1 is perpendicular to the plane BCD , by homothety, GG_1 is also perpendicular to the plane BCD . Since O is the common **circumcentre** of $ABCD$ and $A_1 BCD$, the points N and N_1 lie on OG and OG_1 such that

$$OG : GN = OG_1 : G_1 N_1 = 3 : 1.$$

Therefore, NN_1 is also perpendicular to the plane BCD .



Let the distances from A and A_1 to the plane BCD be d and $-d'$ respectively, where the negative sign means A_1 lie on different sides as A to the plane. Then the distance from O to the plane BCD is $\frac{d-d'}{2}$. By using the homothety, the distances from G, G_1, N, N_1 to the plane BCD are

$$\frac{d}{4}, -\frac{d'}{4}, \frac{1}{(-1)+4} \left(-\frac{d-d'}{2} + 4 \cdot \frac{d}{4} \right) = \frac{d+d'}{6}, \frac{1}{(-1)+4} \left(-\frac{d-d'}{2} - 4 \cdot \frac{d'}{4} \right) = -\frac{d+d'}{6}.$$

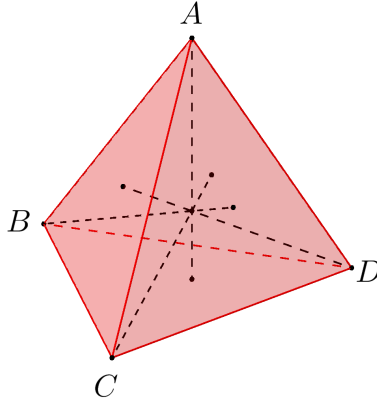
This shows N and N_1 are symmetric points about the plane BCD . It follows from the reflection that the **twelve-point centre** of A_2BCD is N .

Also, the **twelve-point sphere** of A_2BCD has the same radius as that of A_1BCD , which is $\frac{1}{3}$ of the **circumradius** of A_1BCD , i.e. that of $ABCD$. This is equal to the radius of the **twelve-point sphere** of $ABCD$. Therefore, A_2BCD and $ABCD$ have the same **twelve-point sphere**. Thus, the centroids of $\triangle A_2BC$, $\triangle A_2BD$ and $\triangle A_2CD$ lie on the **twelve-point sphere** of $ABCD$.

3 Orthocentric tetrahedron

Given a **tetrahedron** $ABCD$, the *altitude*(高) from A is the line passing through A and perpendicular to the plane BCD . In $\mathbb{R}^2$, the 3 altitudes of a triangle must intersect at the orthocentre of the triangle. However, it is not hard to find some counterexample for which the 4 **altitudes** of $ABCD$ do not intersect at a point. In this section, we investigate a special kind of **tetrahedron** for which the 4 **altitudes** are concurrent.

Definition 3.1. A **tetrahedron** is called an *orthocentric tetrahedron*(垂心四面體) if its 4 **altitudes** are concurrent. The concurrent point is called the *orthocentre*(垂心) of the **tetrahedron**.



Before we give some properties of **orthocentric tetrahedrons**, we first say something about how to prove two lines AB and CD are perpendicular. The most common way is to show that AB is a **normal** to a plane containing CD . For example, if $AB \perp CE$ and $AB \perp DE$ for some point E , then AB must be perpendicular to the plane CDE (or possibly a line), so that $AB \perp CD$. Another way is given as follows.

Proposition 3.1. Let A, B, C, D be distinct points in $\mathbb{R}^3$. Then $AB \perp CD$ iff

$$AC^2 - AD^2 = BC^2 - BD^2.$$

Proof. Let $\mathbf{a}, \mathbf{b}, \mathbf{c}, \mathbf{d}$ be the **position vectors** of A, B, C, D respectively. Then

$$\begin{aligned} AC^2 - AD^2 &= BC^2 - BD^2 \\ \Leftrightarrow \|\mathbf{a} - \mathbf{c}\|^2 - \|\mathbf{a} - \mathbf{d}\|^2 &= \|\mathbf{b} - \mathbf{c}\|^2 - \|\mathbf{b} - \mathbf{d}\|^2 \\ \Leftrightarrow \|\mathbf{c}\|^2 - \|\mathbf{d}\|^2 - 2\mathbf{a} \cdot \mathbf{c} + 2\mathbf{a} \cdot \mathbf{d} &= \|\mathbf{c}\|^2 - \|\mathbf{d}\|^2 - 2\mathbf{b} \cdot \mathbf{c} + 2\mathbf{b} \cdot \mathbf{d} \\ \Leftrightarrow -\mathbf{a} \cdot \mathbf{c} + \mathbf{a} \cdot \mathbf{d} &= -\mathbf{b} \cdot \mathbf{c} + \mathbf{b} \cdot \mathbf{d} \\ \Leftrightarrow (\mathbf{a} - \mathbf{b}) \cdot (\mathbf{d} - \mathbf{c}) &= 0 \\ \Leftrightarrow AB &\perp CD. \end{aligned}$$

□

The following give several equivalent ways to show that a **tetrahedron** is **orthocentric**.

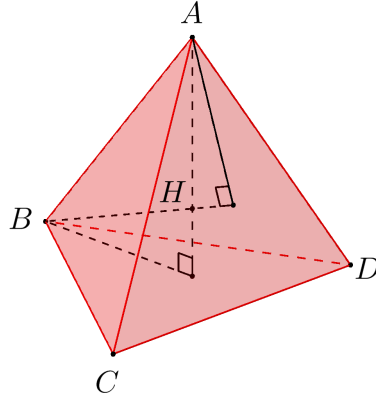
Proposition 3.2. Let $ABCD$ be a **tetrahedron**. Then the following are equivalent.

- (i) $ABCD$ is an **orthocentric tetrahedron**
- (ii) $AB \perp CD$, $AC \perp BD$ and $AD \perp BC$
- (iii) $AB^2 + CD^2 = AC^2 + BD^2 = AD^2 + BC^2$
- (iv) the **circumscribed parallelepiped** of $ABCD$ is a **rhombohedron**
- (v) the 3 **bimedians** of $ABCD$ have the same length

Proof. We prove the following implications.

- (i) $\Rightarrow$ (ii)

Let the **orthocentre** be H . Since AH is perpendicular to the plane BCD , we have $AH \perp CD$. Similarly, $BH \perp CD$. So CD is a **normal** to the plane ABH . Thus, $CD \perp AB$. By symmetry, (ii) holds.

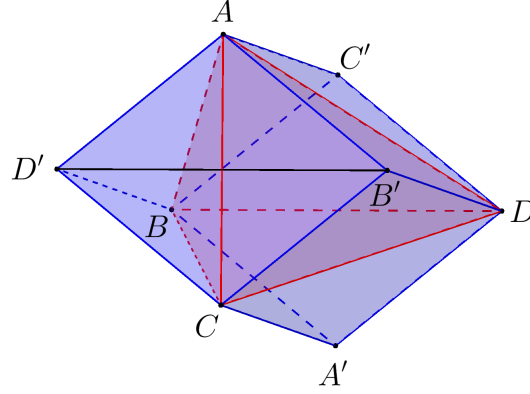


- (ii) $\Leftrightarrow$ (iii)

By proposition 3.1, $AD \perp BC$ iff $AB^2 + CD^2 = AC^2 + BD^2$. Similarly, each perpendicular condition in (ii) is equivalent to the equality of a pair of expressions in (iii). So these statements are equivalent. (Note that this also shows that when two perpendicular conditions in (ii) hold, then the third also holds.)

- (ii) $\Rightarrow$ (iv)

Let A', B', C', D' be the opposite vertices of A, B, C, D in the **circumscribed parallelepiped**. Then $B'D' \parallel BD$, and hence $B'D' \perp AC$. Since $AD'CB'$ is a parallelogram, it is indeed a rhombus. Similarly, all faces of the **circumscribed parallelepiped** is a rhombus, and the **parallelepiped** is a **rhombohedron**.



- (iv) $\Leftrightarrow$ (v)

Recall that the **bimedians** of $ABCD$ have the same length as AB', AC', AD' . So they have the same length iff $AB' = AC' = AD'$ iff $AC'BD'CA'DB'$ is a **rhombohedron**.

- (iv) $\Rightarrow$ (i)

Since $AB' = AC' = AD'$ and the A -**altitude** of $ABCD$ is perpendicular to the plane BCD , hence also to the plane $B'C'D'$, this **altitude** is the locus of all points which are equidistant to B', C', D' . Therefore, the **circumcentre** O' of $A'B'C'D'$ lies on this **altitude**. Similarly, O' lies on all **altitudes** of $ABCD$, and so these **altitudes** are concurrent.

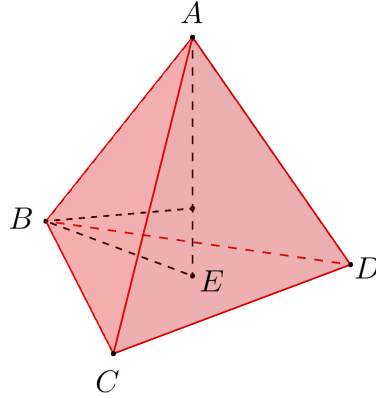
All these together prove the equivalence of the statements. $\square$

From the above proof, we immediately obtain the following fact. The **orthocentre** of an **orthocentric tetrahedron** is the **circumcentre** of the **tetrahedron** formed by the remaining 4 vertices of the **circumscribed parallelepiped**. As a result, the **orthocentre** is simply the **Monge point** of the **orthocentric tetrahedron**. This explains why we use the **Monge point** as an analogue of the orthocentre.

We give one more property of **orthocentric tetrahedrons**.

Proposition 3.3. Let $ABCD$ be an **orthocentric tetrahedron**. Then the foot of **altitude** from A is the orthocentre of $\triangle BCD$.

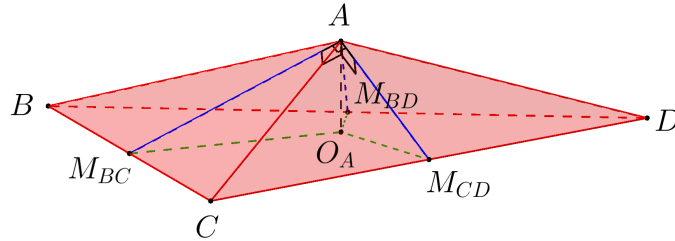
Proof. Let E be the foot of **altitude** from A . Recall from the proof of proposition 3.2 that CD is perpendicular to the plane ABE . So $CD \perp BE$. By symmetry, $BC \perp DE$ and $BD \perp CE$. Thus, E is the orthocentre of $\triangle BCD$ as desired. $\square$



Problem 3.1. Let $ABCD$ be a **tetrahedron**. Suppose the medians from A of $\triangle ABC$, $\triangle ABD$ and $\triangle ACD$ are pairwise perpendicular. Prove that $AB = AC = AD$.

Analysis. This problem is unrelated to **orthocentric tetrahedrons**. It only demonstrates some proof related to perpendicularity. It is very useful to apply the concept of **normals** to planes to interpret the perpendicular conditions.

Solution. Let the midpoints of BC, CD, BD be M_{BC}, M_{CD}, M_{BD} respectively, and let O_A be the projection of A on the plane BCD .



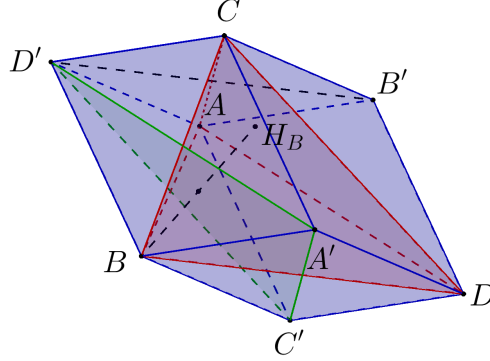
As AM_{BC} is perpendicular to both AM_{CD} and AM_{BD} , it is perpendicular to the plane $AM_{CD}M_{BD}$. So it is perpendicular to $M_{BD}M_{CD}$, and hence to BC by the midpoint theorem. Therefore, the projection $O_A M_{BC}$ is perpendicular to BC as well. This yields $O_A B = O_A C$, and thus $AB = AC$. Similarly, $AB = AC = AD$.

Problem 3.2. Let $ABCD$ be an **orthocentric tetrahedron**. Let H_B, H_C, H_D be the feet of **altitudes** from B, C, D respectively. Prove that B, C, D, H_B, H_C, H_D are **cospherical**.

Analysis. To solve this problem, we need one more observation. The **mediator** of the **altitude** $\overline{BH_B}$ is the plane $A'C'D'$, where A', C', D' are vertices of the **circumscribed parallelepiped**. Once this is obtained, it is not hard to show that A' is equidistant to B, C, D, H_B, H_C, H_D , and hence the 6 points are **cospherical**.

Solution. Consider the **circumscribed parallelepiped**, and let A', B', C', D' be the opposite vertices to A, B, C, D respectively. Recall from proposition 2.2(b) that the

planes $A'C'D'$ and ACD trisect BB' . Therefore, the point B and the plane ACD have the same distance to the plane $A'C'D'$. This implies the plane $A'C'D'$ is the **mediator** of BH_B . Thus, $A'B = A'H_B$.



By symmetry, we also have $A'C = A'H_C$ and $A'D = A'H_D$. In addition, since $ABCD$ is **orthocentric**, its **circumscribed parallelepiped** is a **rhombohedron** by proposition 3.2. This yields $A'B = A'C = A'D$. Hence, A' is equidistant to B, C, D, H_B, H_C, H_D , and these 6 points lie on a **sphere** with centre A' and radius $A'B$ (which is the length of a **bimedian** of $ABCD$).

Links

Theorems

- [Carnot's theorem](#)
- [Cauchy-Schwarz inequality](#)
- [Classification of Platonic solids](#)
- [Commandino's theorem](#)
- [Cosine formula for tetrahedrons](#)
- [Descartes' theorem](#)
- [Euler's polyhedron formula](#)
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Propositions

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- [2.3: Analogue of the angle bisector theorem](#)
- [3.1: Generalization of Pythagoras' theorem in \$\mathbb{R}^3\$](#)
- [3.2: Criteria for a tetrahedron to be orthocentric](#)
- [3.3: Feet of altitudes of orthocentric tetrahedron](#)

Problems

- [1.1: Dot product and angle](#)

- 1.2: Sharygin Geometry Olympiad 2011 Final Grade 10 Problem 3
- 1.3: Dihedral angle
- 1.4: Analogue of the intercept theorem
- 1.5: Carnot's theorem
- 1.6: Power of point with respect to sphere
- 1.7: USAMO 1974 Problem 3 modified
- 1.8: Proof of proposition 1.6
- 2.1: IMO 1964 Problem 6 part 1
- 2.2: In the World of Mathematics Problem 296
- 2.3: All-Siberian Open School Olympiad 2013 Grade 11 Problem 4
- 2.4: Analogue of the Miquel point
- 2.5: IMO Longlist 1985
- 2.6: Sharygin Geometry Olympiad 2020 Correspondence Problem 24
- 2.7: Tangential spheres
- 2.8: Twelve-point spheres of tetrahedrons
- 3.1: Perpendicular medians
- 3.2: Cospherical points in orthocentric tetrahedron

Terminologies

- altitude
- angular defect
- antipode
- bimedian
- bisecting plane
- Cayley-Menger determinant
- centre (sphere)
- centroid
- circumcentre
- circumradius
- circumscribed parallelepiped
- circumsphere
- cone

- convex polyhedron
- coplanar
- cospherical
- cross product $\mathbf{u} \times \mathbf{v}$
- cube
- cuboid
- cylinder
- diameter
- dihedral angle
- dot product $\mathbf{u} \cdot \mathbf{v}$
- edge
- Euclidean space $\mathbb{R}^n$
- Euler line
- excentre
- exsphere
- external bisecting plane
- face
- great circle
- hemisphere
- incentre
- inradius
- insphere
- internal bisecting plane
- median
- mediating plane
- mediator
- Monge plane
- Monge point
- norm $\|\mathbf{v}\|$
- normal (vector)
- orthocentre
- orthocentric tetrahedron

- parallelepiped
- Platonic solid
- polyhedron
- position vector $\overrightarrow{OP}$
- power
- prism
- pyramid
- radius
- regular dodecahedron
- regular icosahedron
- regular octahedron
- regular polyhedron
- rhombohedron
- right pyramid
- scalar
- scalar multiplication
- skew lines
- small circle
- sphere
- standard vector $\mathbf{i}, \mathbf{j}, \mathbf{k}$
- stereographic projection
- tangent
- tetrahedron
- twelve-point centre
- twelve-point sphere
- unit vector
- vector
- vertex
- zero vector

References

- Modern Pure Solid Geometry (by Nathan Altshiller Court)
- Wikipedia — Tetrahedron (available from: <https://en.wikipedia.org/wiki/Tetrahedron>)

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